

Rapid Microstructure Prediction in Laser Powder Bed Fusion Using Bayesian Updating of Eagar–Tsai Model

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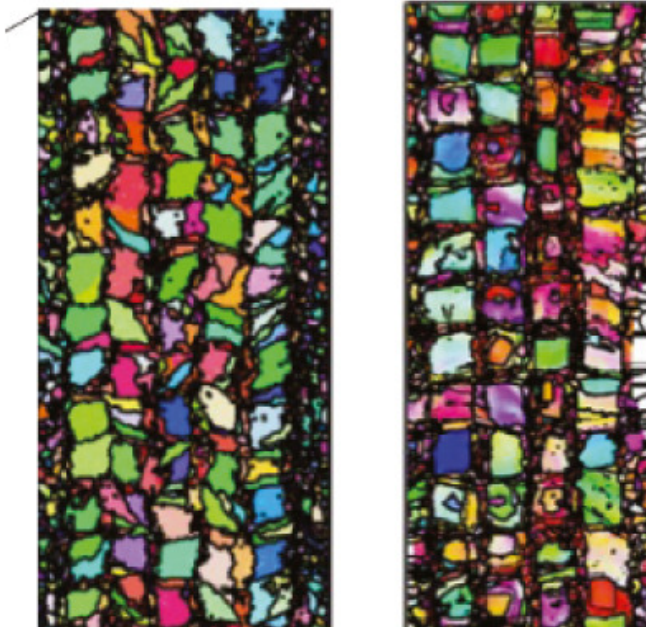
Why Predict Microstructure?

Process parameters affect performance through the thermal history and resulting microstructure.

Composition + Process → Thermal History → Microstructure

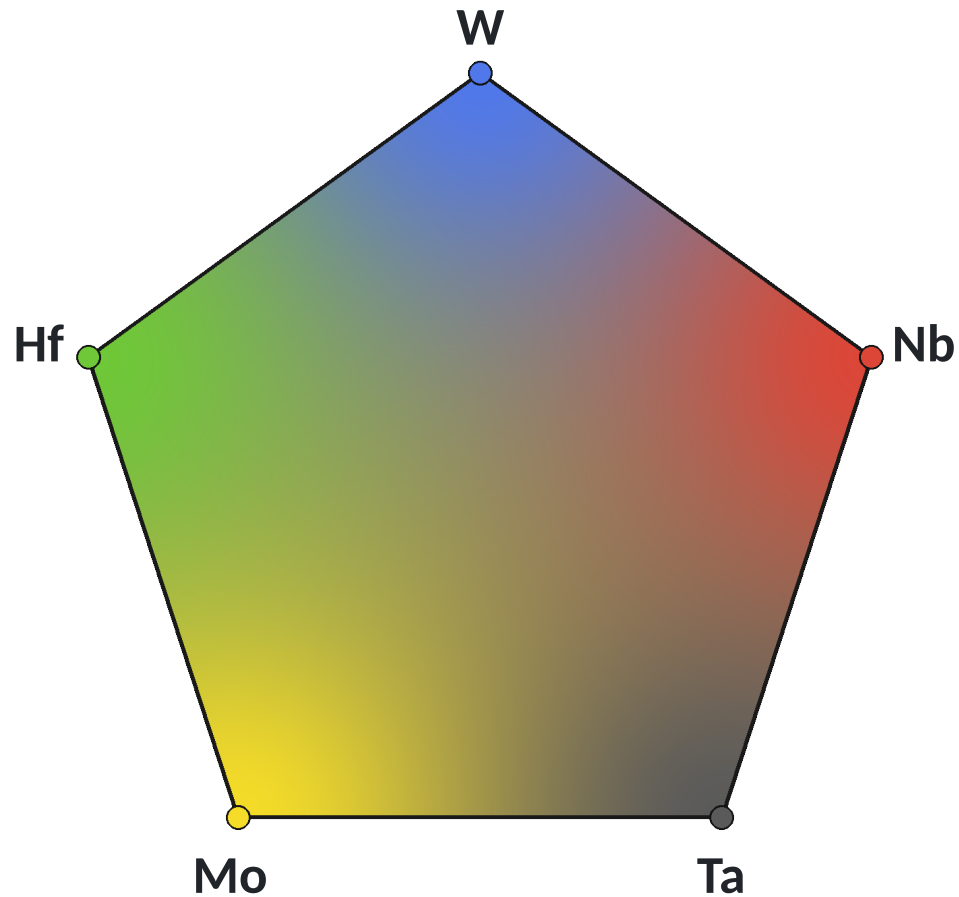
- Microstructure controls strength, ductility, anisotropy, and defect sensitivity.
- A sound melt pool can still solidify into the wrong morphology.
- Predicting structure reduces trial-and-error printing.
- Fast predictions enable composition and process screening.

Simulation Experiment



ExaCA CA prediction (left) and AM-Bench EBSD (right).
M. Rolchigo et al., *Comput. Mater. Sci.* 214 (2022) 111692, Fig. 11(b).

The Design Space Is Large



Five-element projection of a multicomponent composition space.

Each alloy has many possible process conditions

Composition–Process Space

$$\text{Composition} \times P \times v \times \dots$$

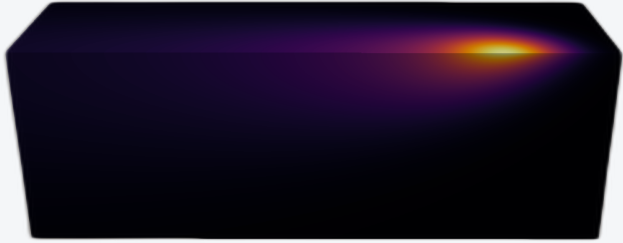
- Cannot practically evaluate every combination with high-fidelity calculations.
- Rapid thermal prediction is needed for high-throughput screening.



Core Thesis

ET is fast enough for high-throughput screening, but not yet reliable enough for microstructure decisions.

Eagar-Tsai



- Fast, but has simplified physics
- Run across the design space

Thermo-Calc AM Module



- Higher fidelity, but more computationally expensive
- Run at selected conditions

Bayesian-Updated ET



- Learn the TCAM-ET discrepancy
- Correct the ET temperature field

+

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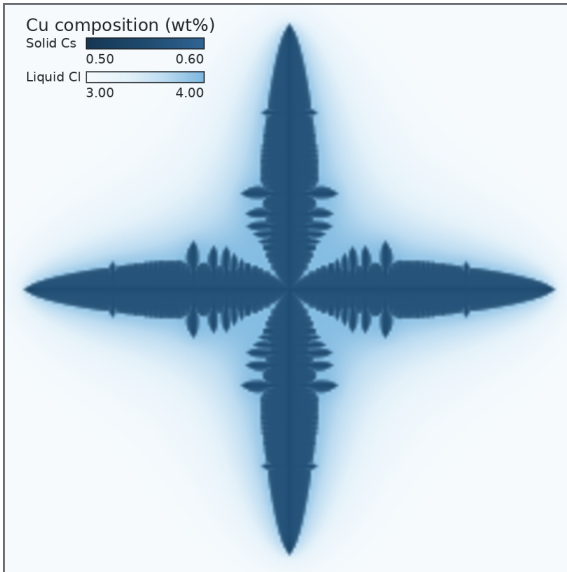
Microstructure Prediction

Corrected 3D temperature field → G and R → KGT → As-solidified microstructure

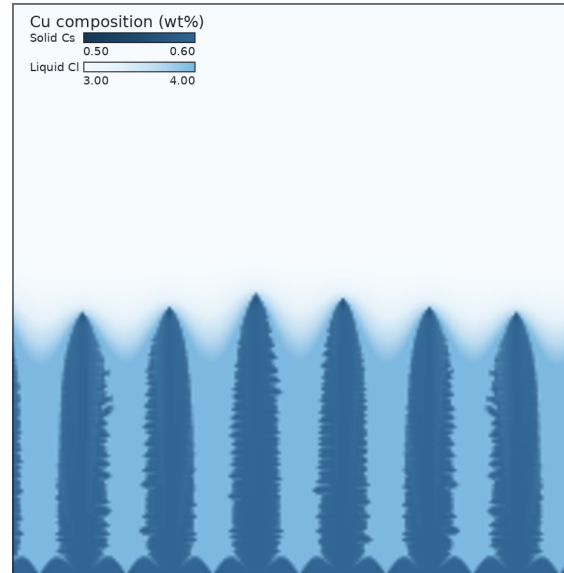


Different Solidification Modes

Equiaxed

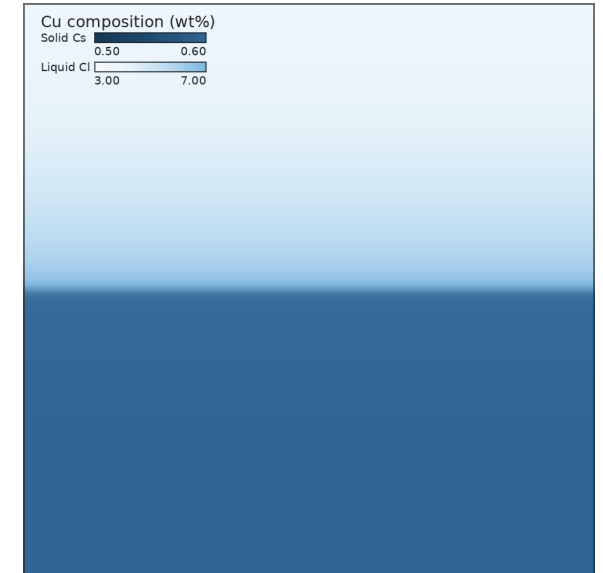


Columnar



Demonstrative cellular automata simulations only.

Planar



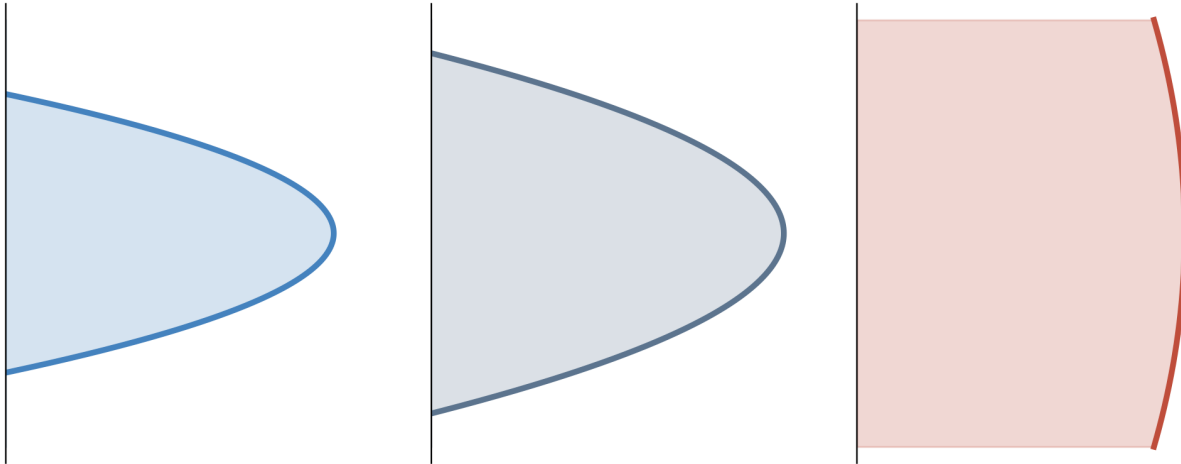
Rapid Microstructural Prediction is the Goal

Eagar–Tsai supplies fast G and R fields; KGT solidification theory and the Gäumann criterion are used to classify the resulting microstructure as planar, columnar, or equiaxed.

KGT Solidification Theory

Kurz-Giovanola-Trivedi framework

- KGT links dendrite-tip operating state to local thermal gradient, G , and interface growth rate, R .
- The dendrite tip is approximated locally as a paraboloid with radius of curvature ρ .
- Tip selection then relates ρ to R and alloy transport and capillarity properties.



Tip relations

$$z = \frac{x^2 + y^2}{2\rho}$$

$$Pe = \frac{\rho R}{2D_l}$$

$$\xi_c = 1 - \frac{2k}{\left[1 + (2\pi/Pe)^2\right]^{1/2} - 1 + 2k}$$

$$\rho = 2\pi \left(\frac{\Gamma}{mG_c\xi_c - G} \right)^{1/2}$$

- x, y : transverse tip coordinates; z : growth-axis coordinate; Pe : solutal Péclet number; D_l : liquid diffusivity; k : partition coefficient; ξ_c : marginal-stability function; Γ : Gibbs-Thomson coefficient; m : liquidus slope; G_c : liquid concentration gradient at the interface.
- The coupled relations determine the selected tip radius ρ from local G , R , and alloy properties.

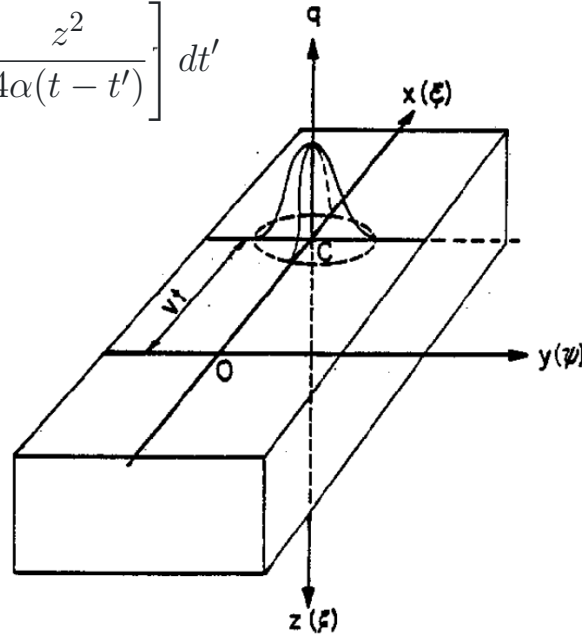


Eagar-Tsai Model Foundation

Original Eagar-Tsai Paper

Transient Gaussian moving heat source

$$\begin{aligned} T - T_0 &= \frac{1}{2} \int_0^t dT_i \\ &= \frac{q}{\pi \rho C_p \sqrt{4\pi\alpha}} \int_0^t \frac{(t-t')^{-1/2}}{2\alpha(t-t') + \sigma^2} \\ &\quad \times \exp \left[-\frac{(x-vt')^2 + y^2}{4\alpha(t-t') + 2\sigma^2} - \frac{z^2}{4\alpha(t-t')} \right] dt' \end{aligned}$$



Inputs

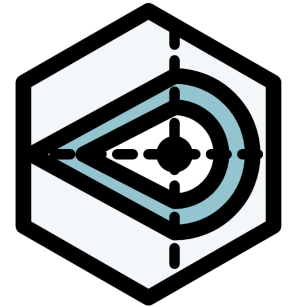
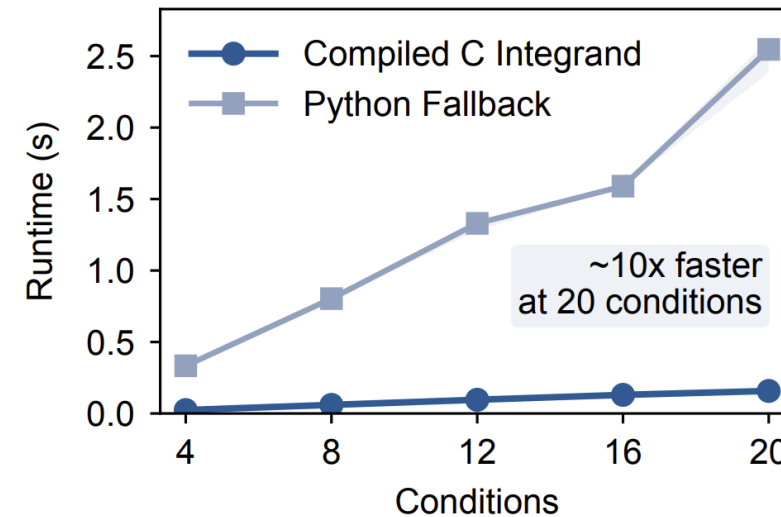
Process: P, v, d_b, A

Material: T_L, k_L, ρ_0, C_p

Eagar, T. W., & Tsai, N. S. (1983). Temperature fields produced by traveling distributed heat sources. *Welding Journal*, 62(12), 346-355.

Refactored Eagar-Tsai Model

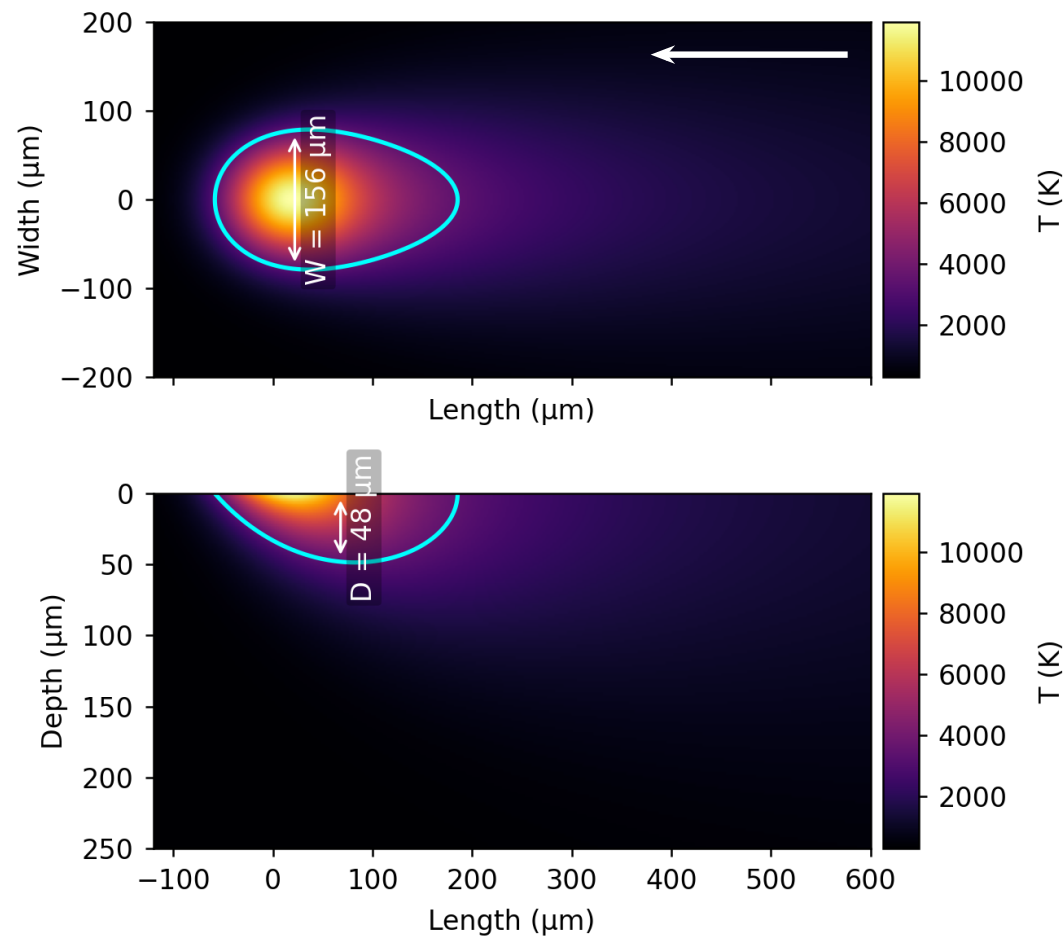
Python library implementation with compiled-C speedup



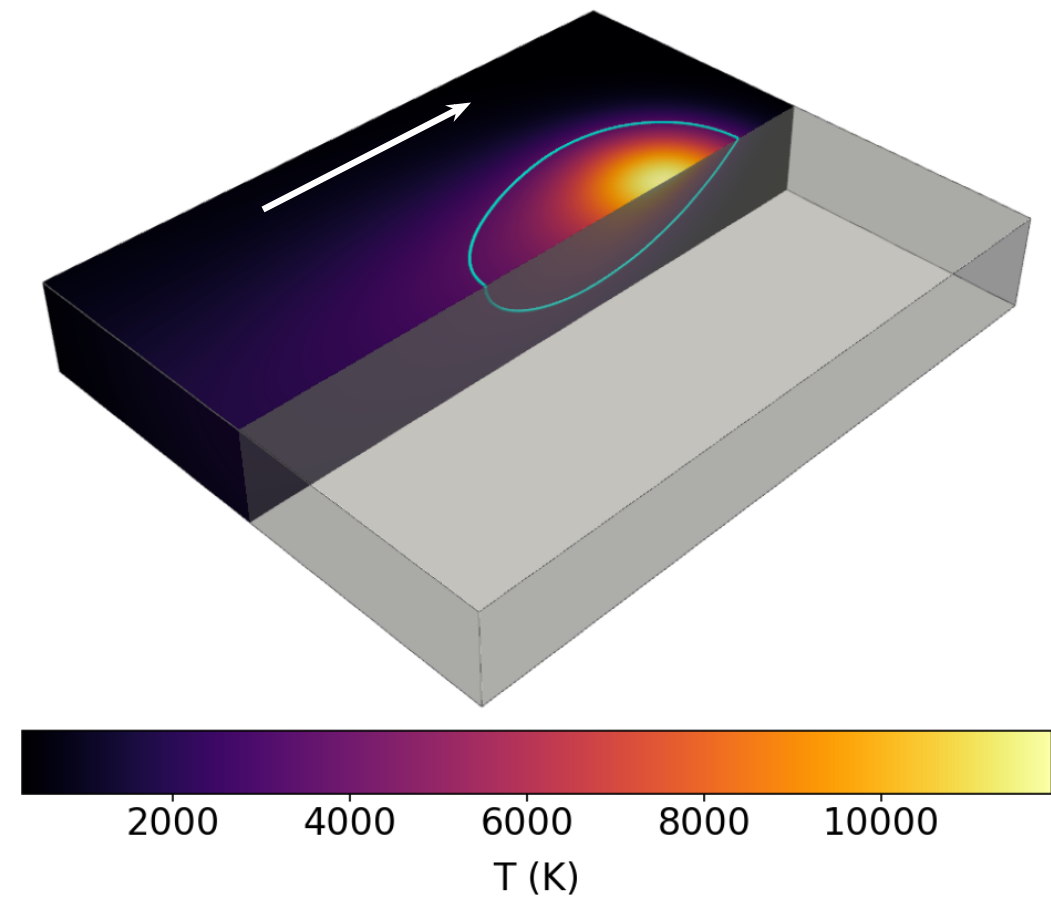
- Separates melt-pool prediction, printability maps, 2D heatmap generation, and 3D temperature-field export into reusable steps.
- Compiled C integrand removes Python overhead inside QUADPACK for faster parameter sweeps.



ET Thermal Cross Sections



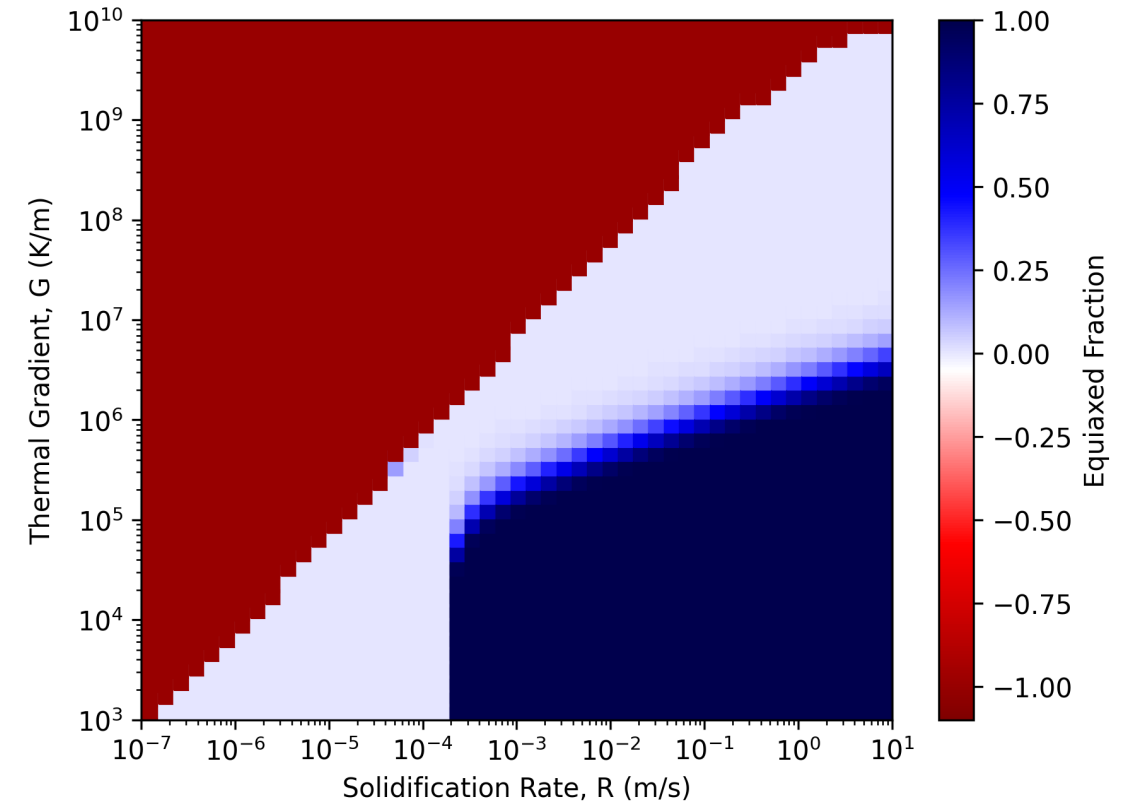
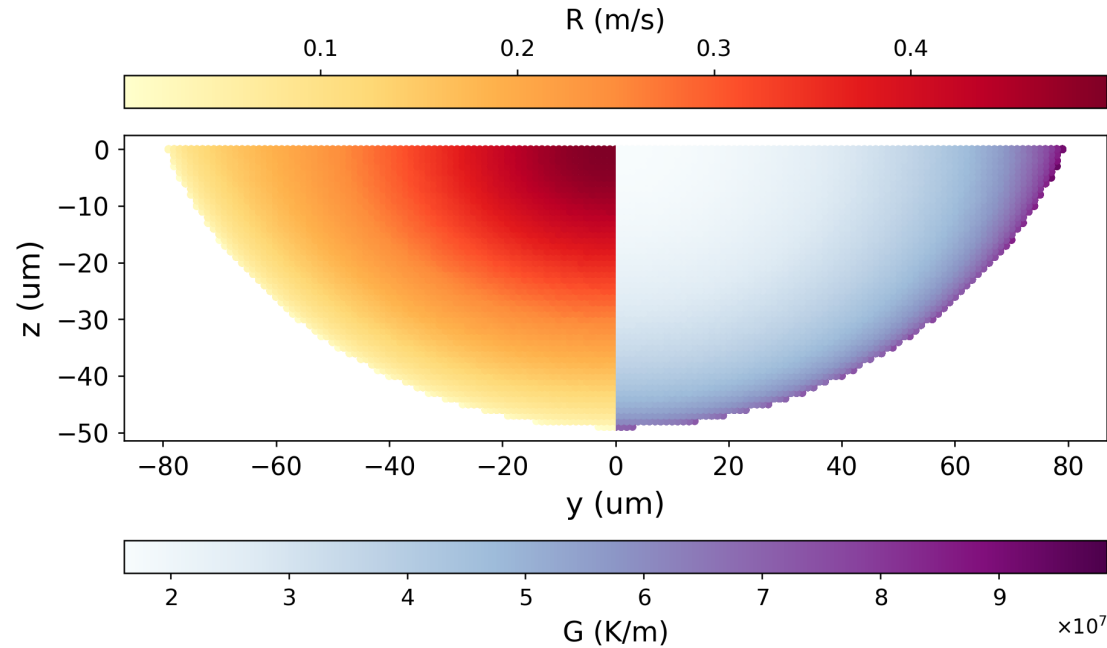
Temperature heatmap for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250 \text{ W}$ and $V = 0.5 \text{ m/s}$. Top: $z = 0$ surface. Bottom: $y = 0$ cross section.



3D view of the ET melt pool for the same alloy and process condition.



ET Thermal Gradient and Solidification Rate

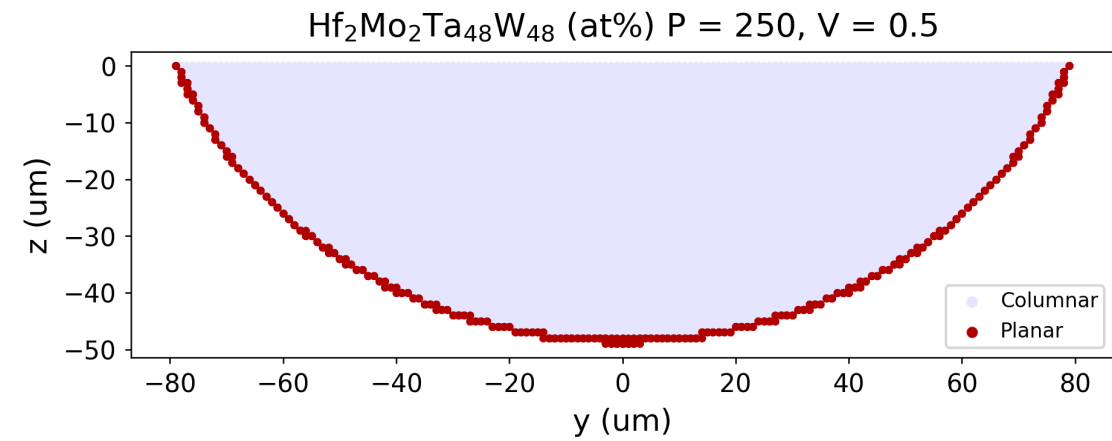
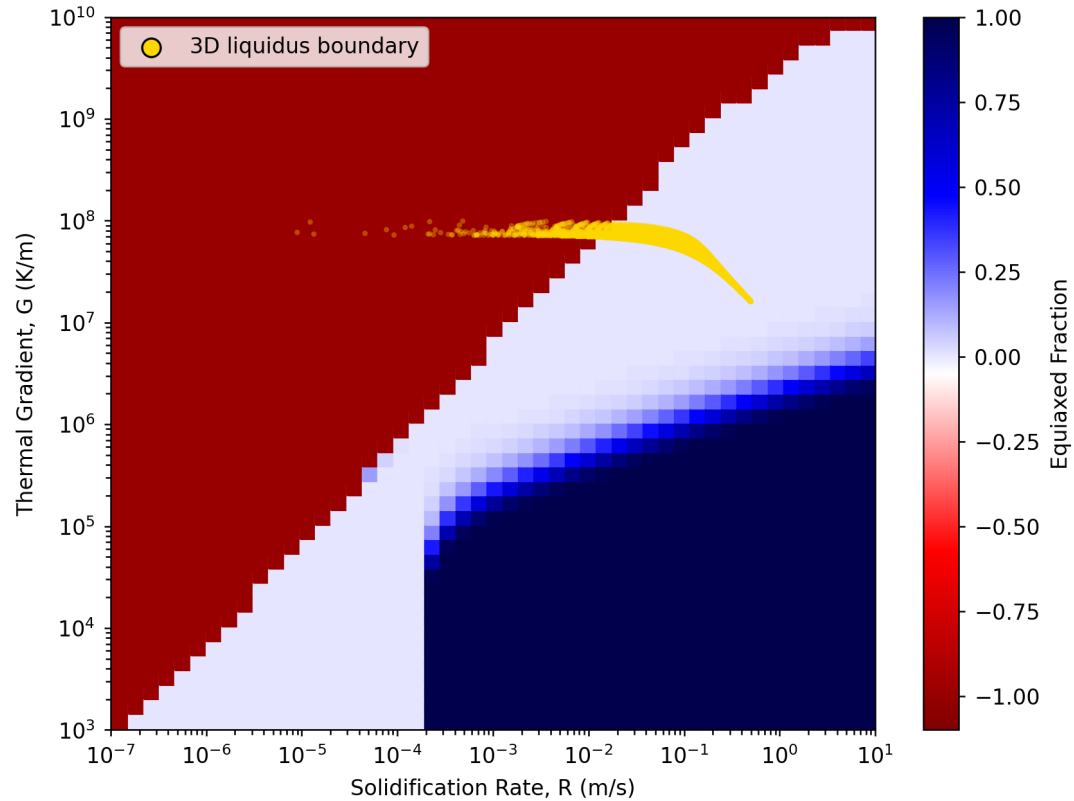


Projection of G and R along the solidification front onto a yz slice for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

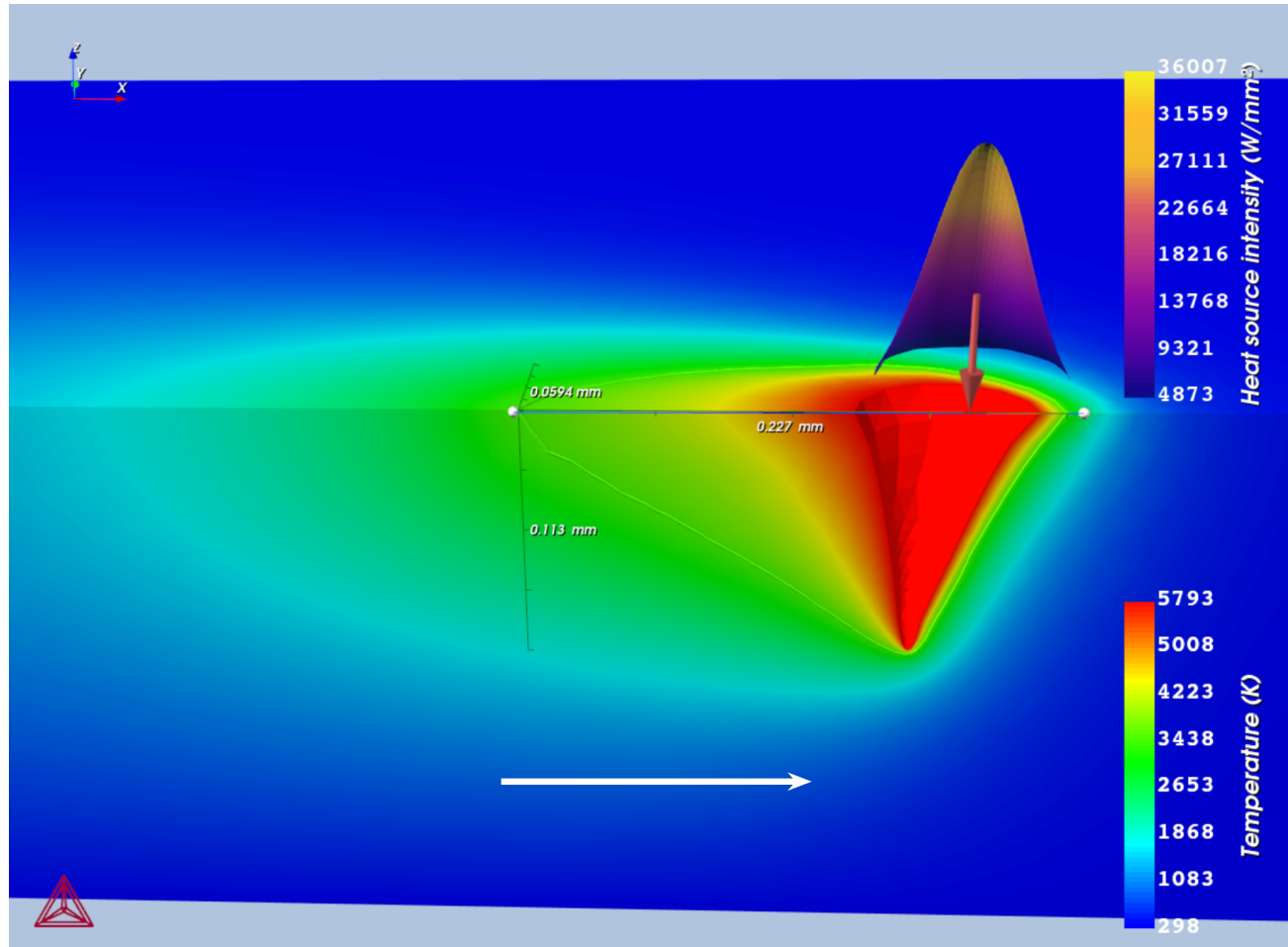
GR Map calculated with Thermo-Calc CET Property Model for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.



ET Microstructure Classification



Thermo-Calc AM Module



Steady-state TCAM simulation for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

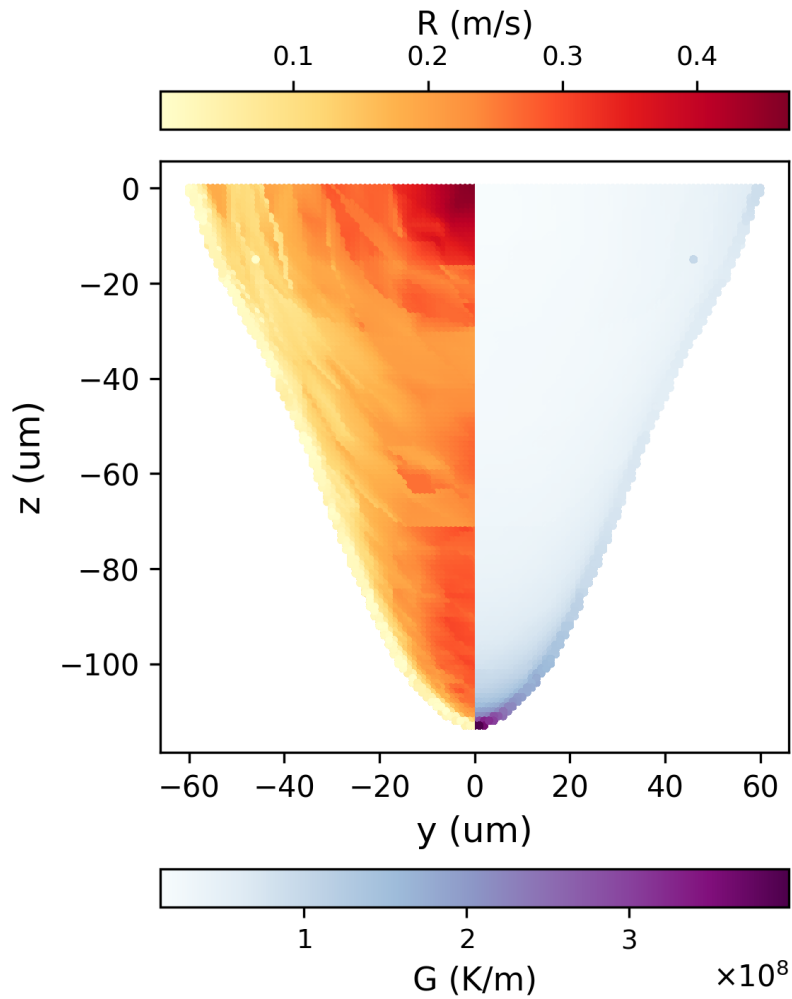
Higher-Fidelity Thermal Reference

- Integrates FEM for thermal conduction with CFD for melt-pool fluid flow.
- Composition-specific, temperature-dependent thermophysical properties.
- Captures Marangoni flow, keyholing, evaporation, radiation, and convection.

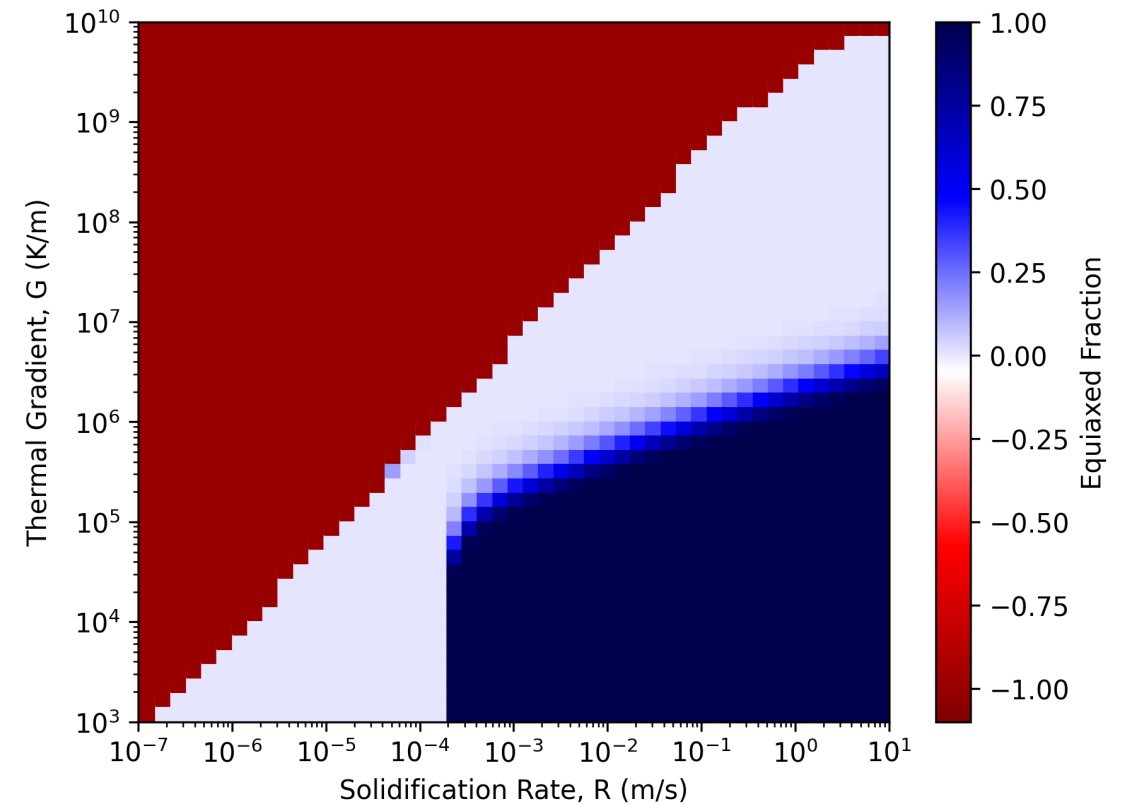
Outputs Used in This Work

- 3D temperature field
- Melt-pool geometry
- Thermal gradient (G) and solidification rate (R) from ParaView post-processing

TCAM Thermal Gradient and Solidification Rate



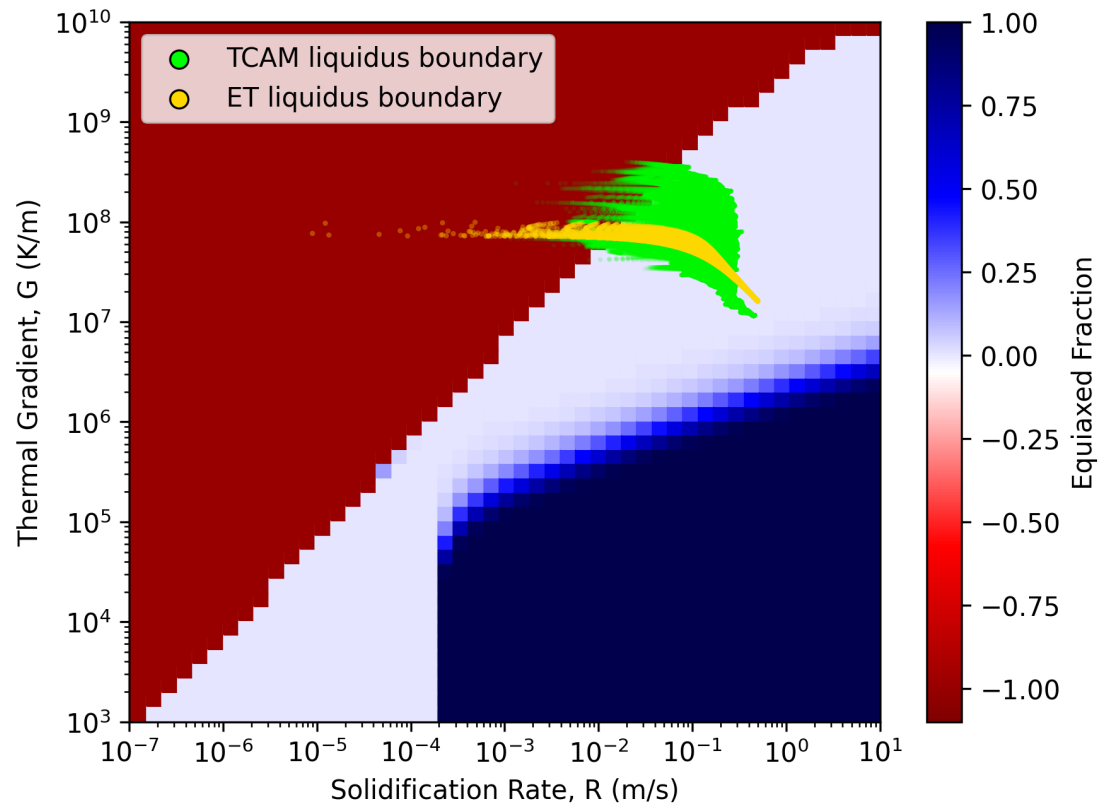
TCAM projection of G and R along the solidification front onto a yz slice for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.



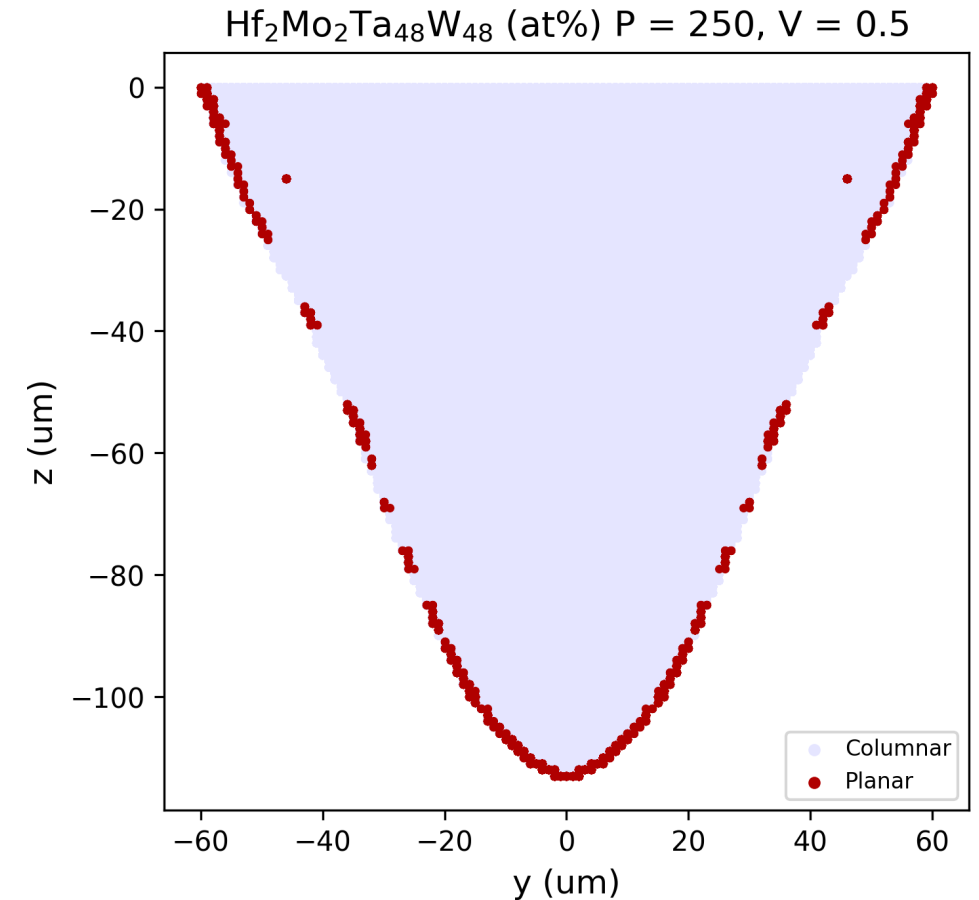
GR Map calculated with the Thermo-Calc CET Property Model for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.



TCAM Microstructure Classification

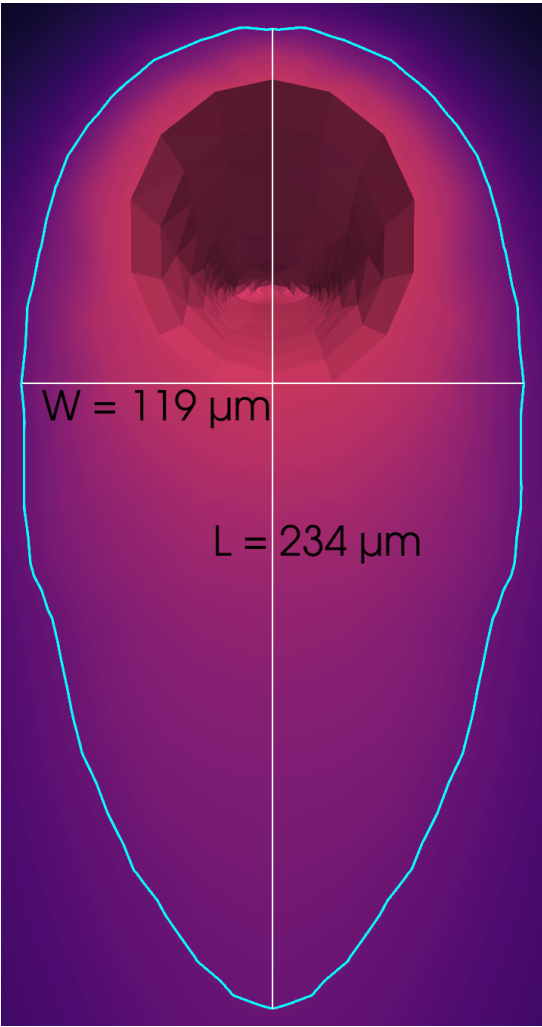
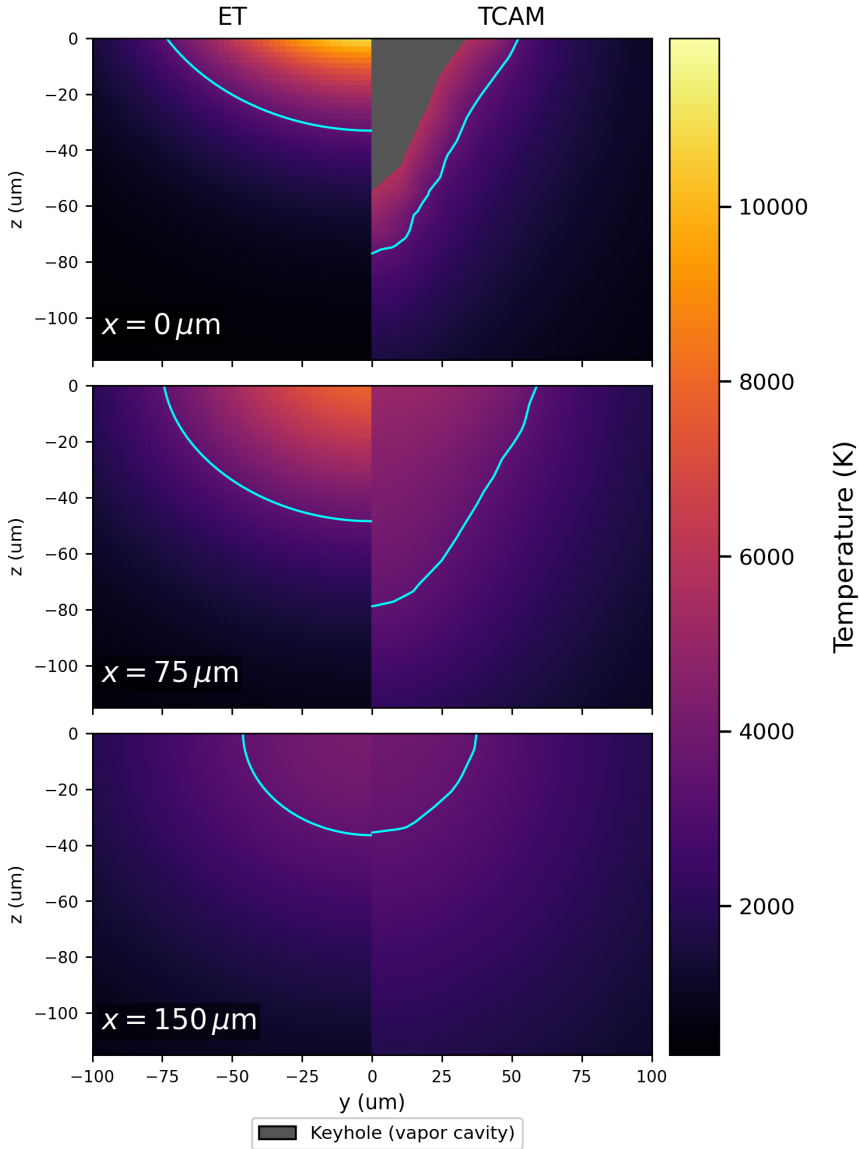
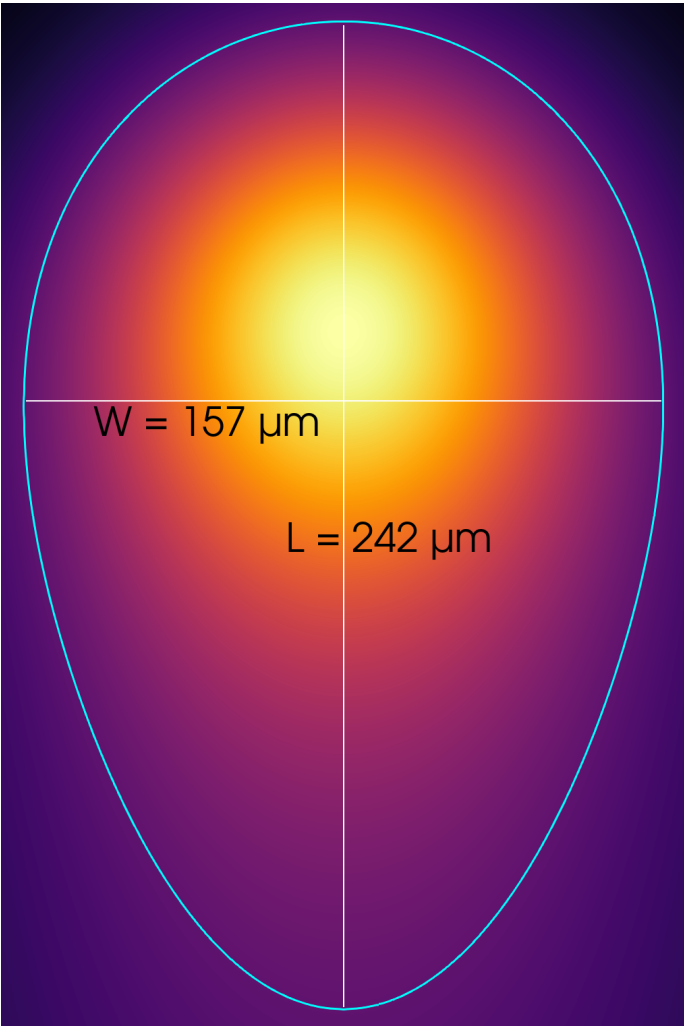


GR map with TCAM and Eagar-Tsai liquidus-boundary conditions overlaid for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.



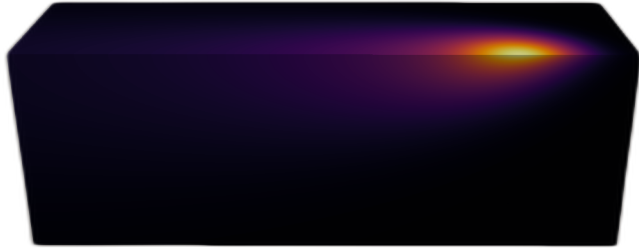
TCAM-derived microstructure classification projected along the solidification front onto a yz slice for the same alloy and process condition.

ET vs. TCAM – Melt Pool Comparison

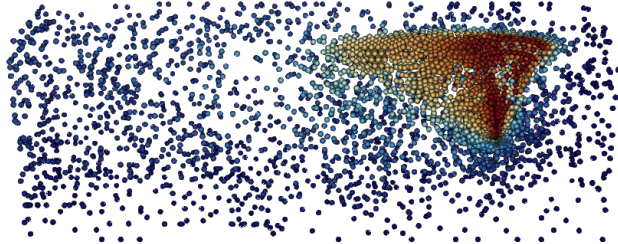


Bayesian Updating

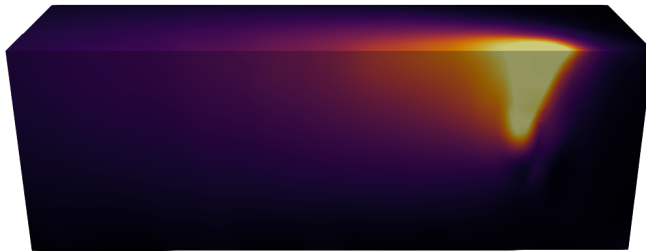
Eagar-Tsai prior



TCAM ground-truth samples



Bayesian-updated ET



1:1 Training Data Reproduction

$$\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48} \quad P = 250 \text{ W} \quad v = 0.5 \text{ m/s}$$

1. ET Prior

Predict the full temperature field rapidly for the selected alloy and process condition.

$$T_{\text{ET}}(x, y, z)$$

2. TCAM Ground-Truth

Compare ET and TCAM temperatures at matched spatial points, then train the Gaussian process on their discrepancy.

$$\Delta T(x, y, z) = T_{\text{TCAM}}(x, y, z) - T_{\text{ET}}(x, y, z)$$

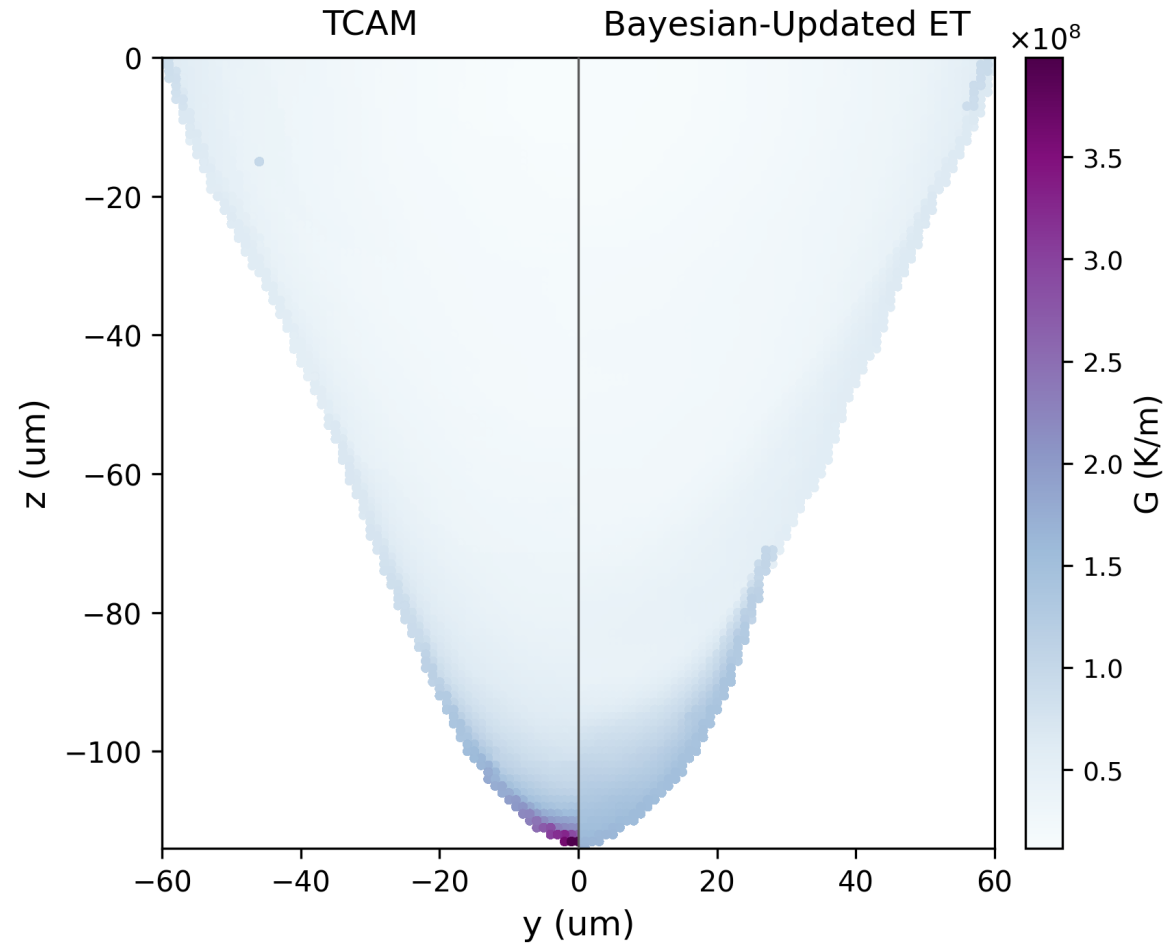
3. Melt Pool Prediction

Add the learned posterior correction to the ET prior, retaining its inexpensive structure while learning the deeper TCAM geometry.

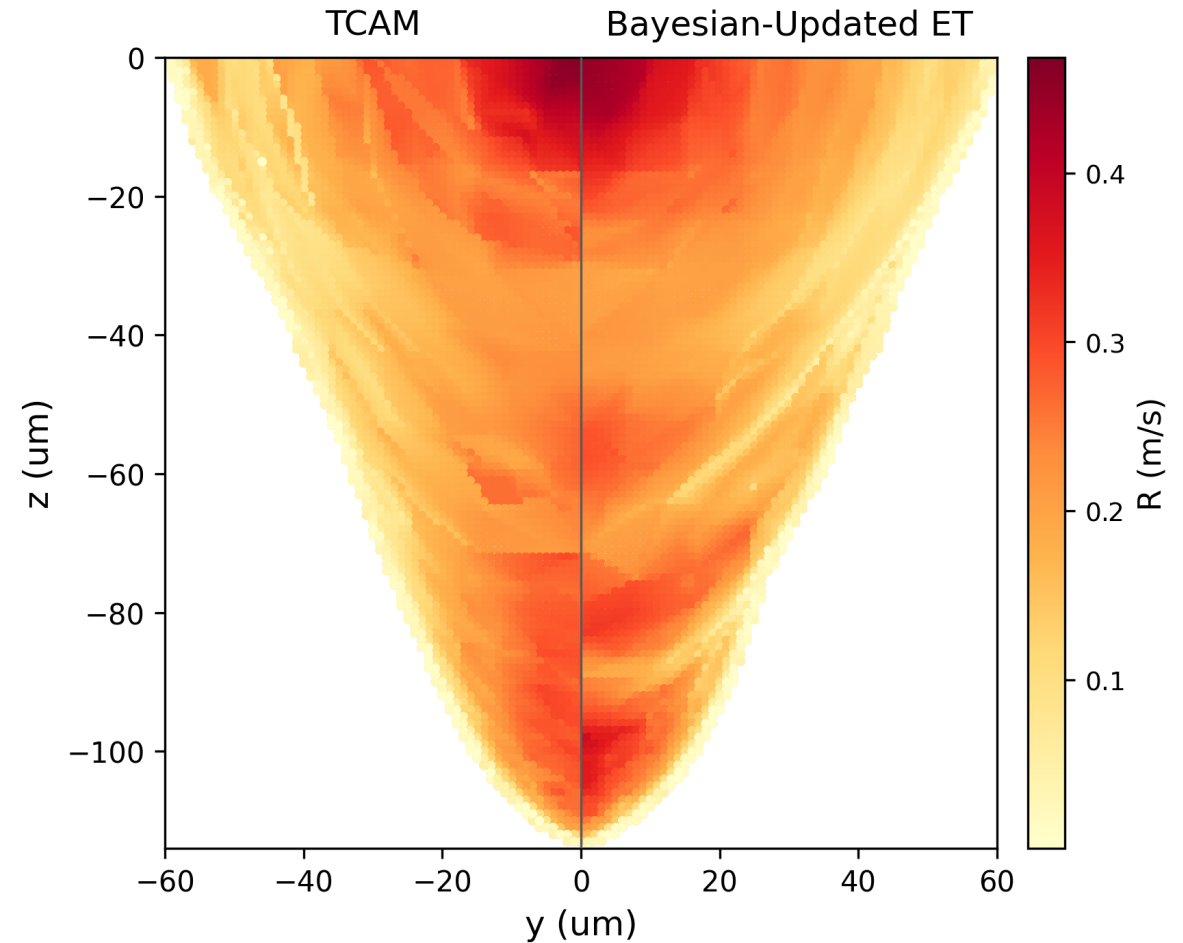
$$T_{\text{updated}}(x, y, z) = T_{\text{ET}}(x, y, z) + \mu_{\Delta T}(x, y, z)$$

Bayesian-Updated ET – Thermal Gradient and Solidification Rate

1:1 Reproducing Training Data: $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$, 250 W, 0.5 m/s



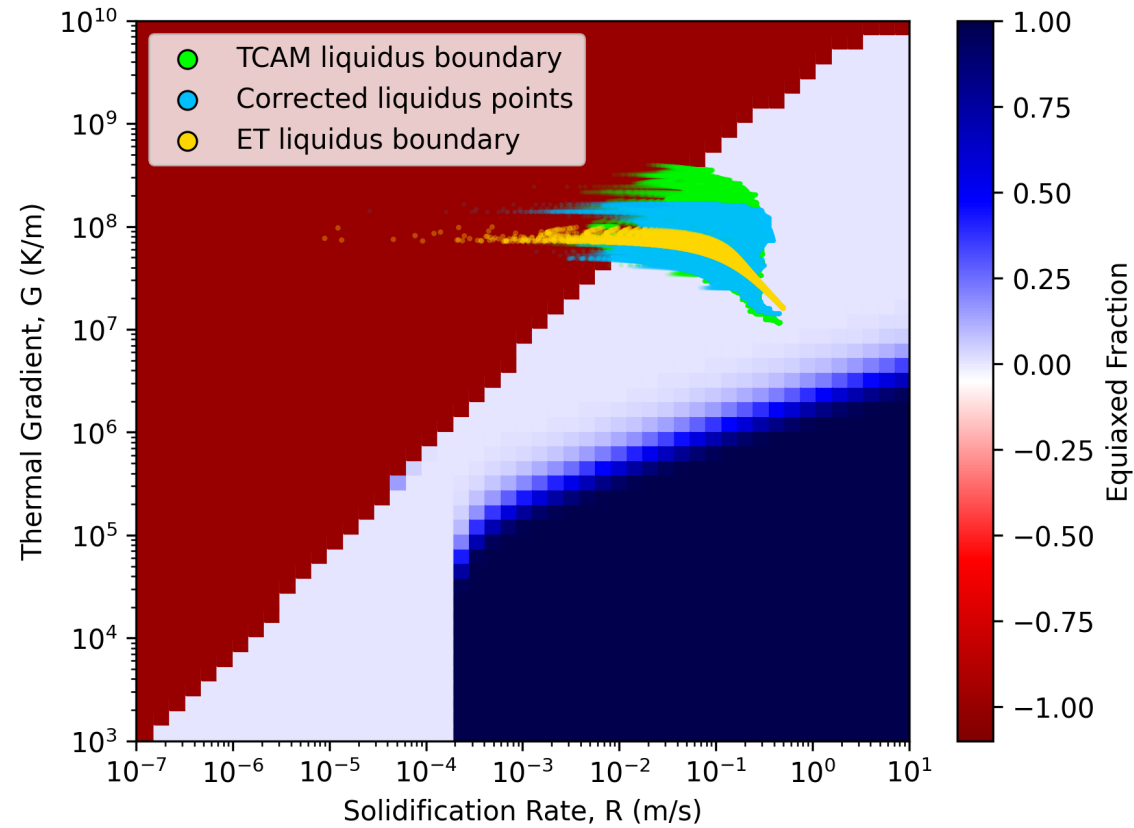
Thermal gradient G along the liquidus solidification front projected onto a yz slice: TCAM (left) and Bayesian-updated ET (right).



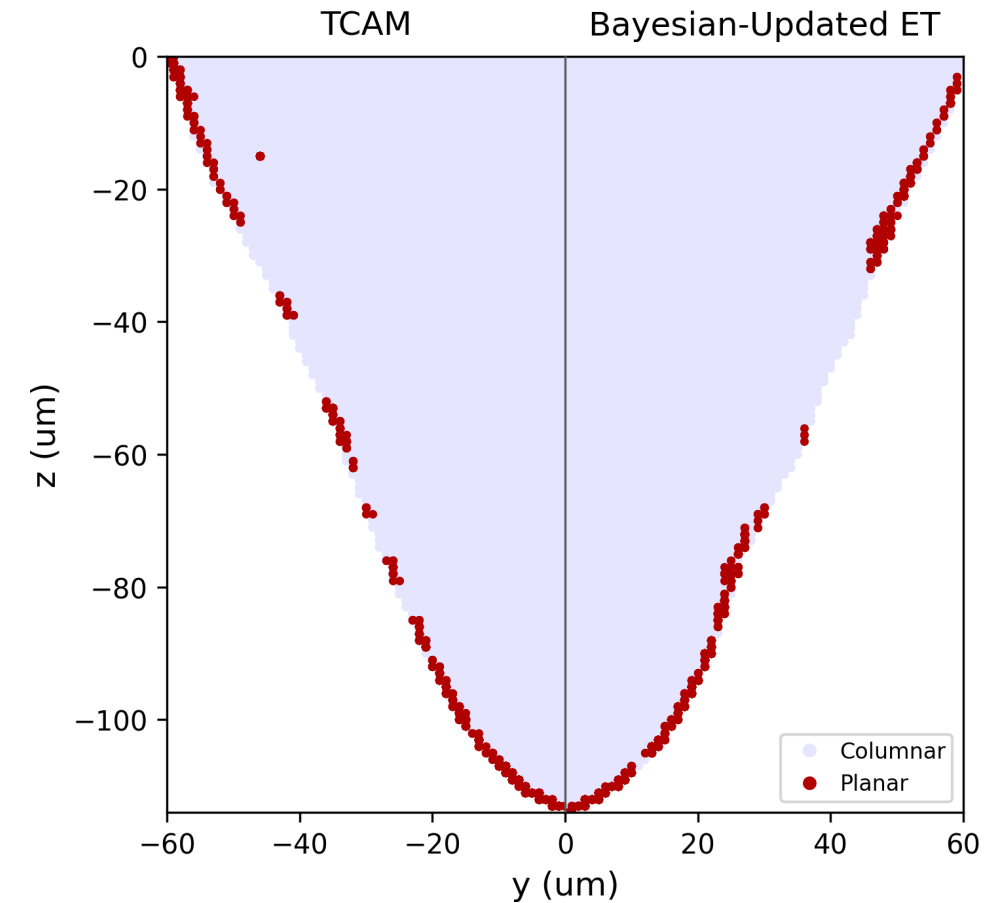
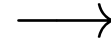
Solidification rate R along the liquidus solidification front projected onto a yz slice: TCAM (left) and Bayesian-updated ET (right).



Bayesian-Updated ET – Microstructure Prediction



Thermo-Calc CET $G-R$ map with TCAM, Bayesian-updated ET, and original ET liquidus-boundary conditions for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$, 250 W and 0.5 m/s.



Microstructure classification along the liquidus solidification front projected onto a yz slice: TCAM (left) and Bayesian-updated ET (right) for the same alloy and process condition.



Warped Multi-Case Gaussian Process

1. Warp the ET Prior

Learn smooth, process-dependent corrections to ET amplitude, front and rear length, width, depth, and x position. The six warp parameters are predicted from power and velocity.

2. Learn the Remaining Residual

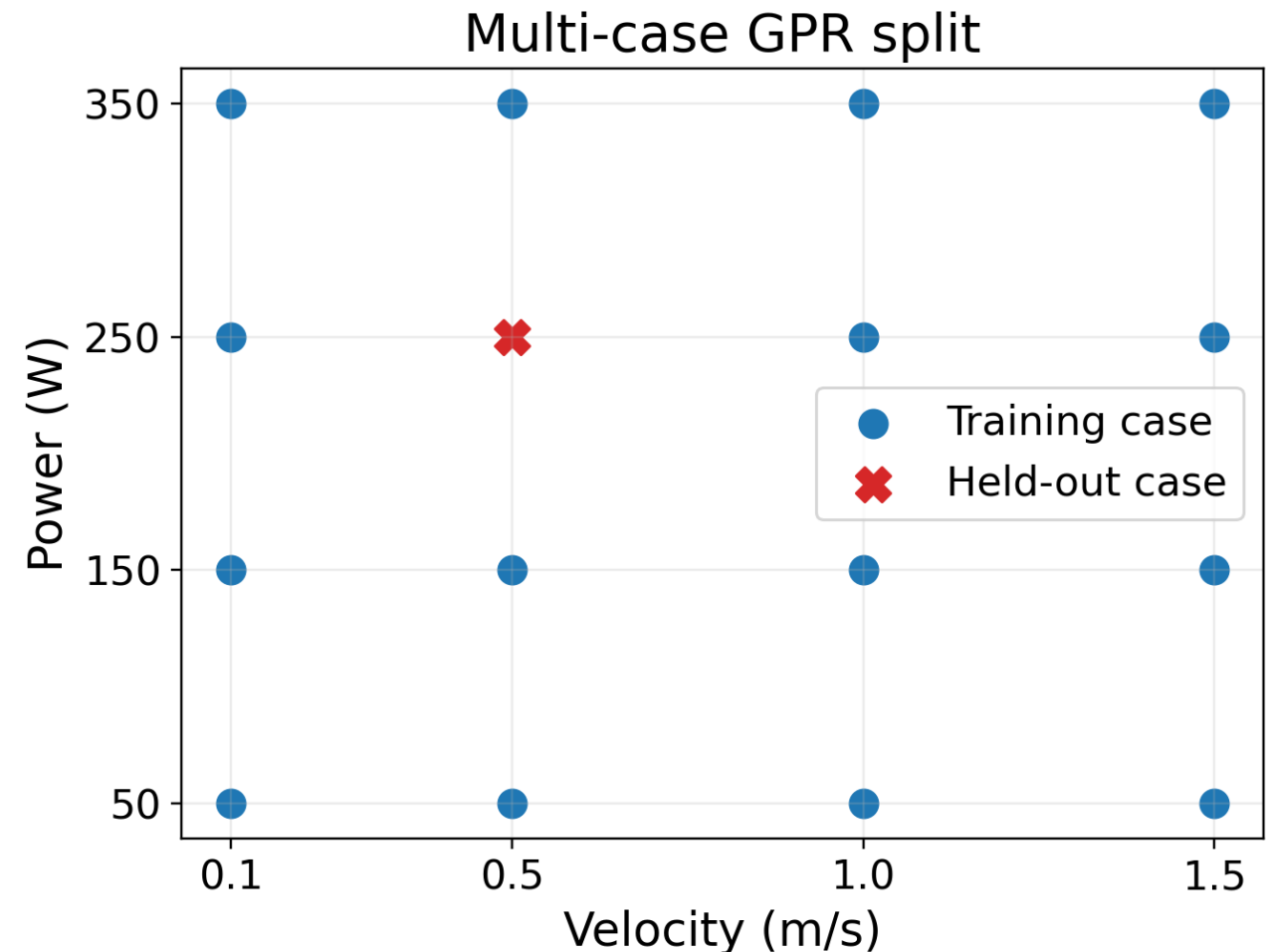
Fit the remaining TCAM-warped-ET discrepancy using an anisotropic **Matérn-5/2 kernel** with one length scale per input and a white-noise term.

GPR Inputs

$$P, v, x, y, z, r_{3D}, T_{\text{warped ET}}$$

$$r_{3D} = \sqrt{x^2 + y^2 + z^2}; \text{ all inputs are normalized before fitting.}$$

15 training cases $\times$ 500 points = 7,500 training points.



Held out: $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$, 250 W, 0.5 m/s

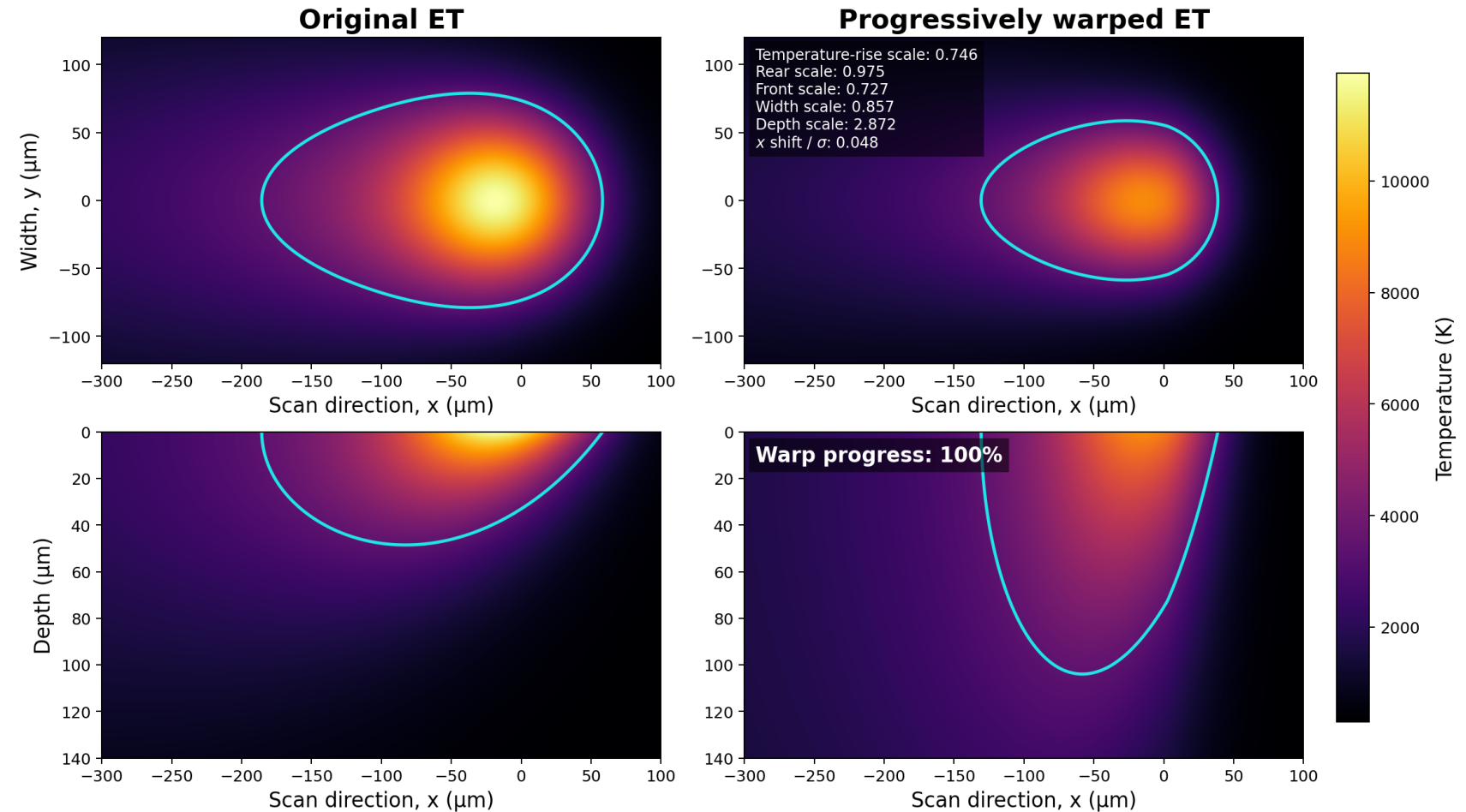


Warped ET Prior

Predicted Warp Parameters

Temperature-rise scale	0.746
Rear scale	0.975
Front scale	0.727
Width scale	0.857
Depth scale	2.872
x shift / σ	0.0476

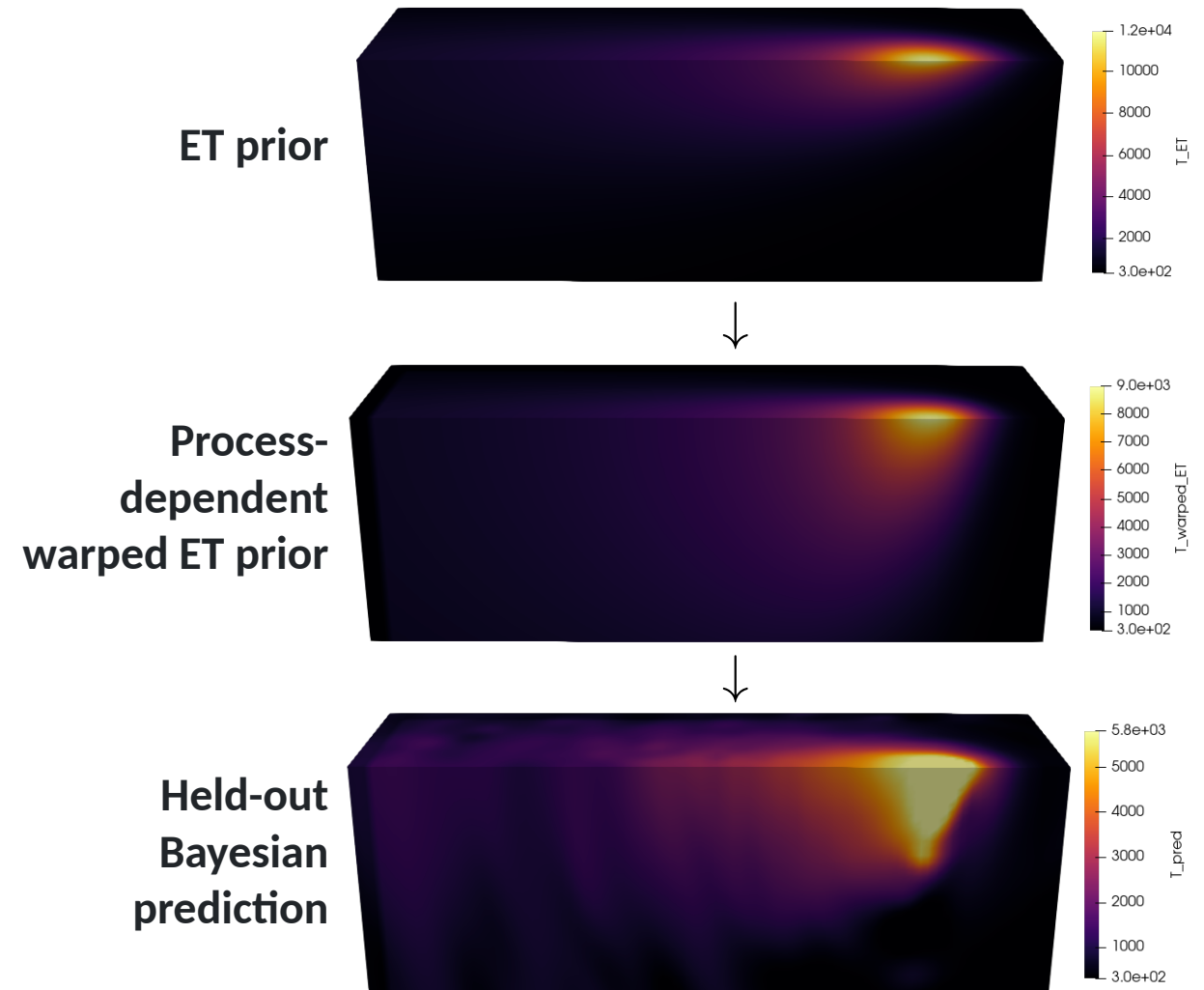
The warp rescales ET temperature amplitude and melt-pool geometry before the residual GPR correction is applied.



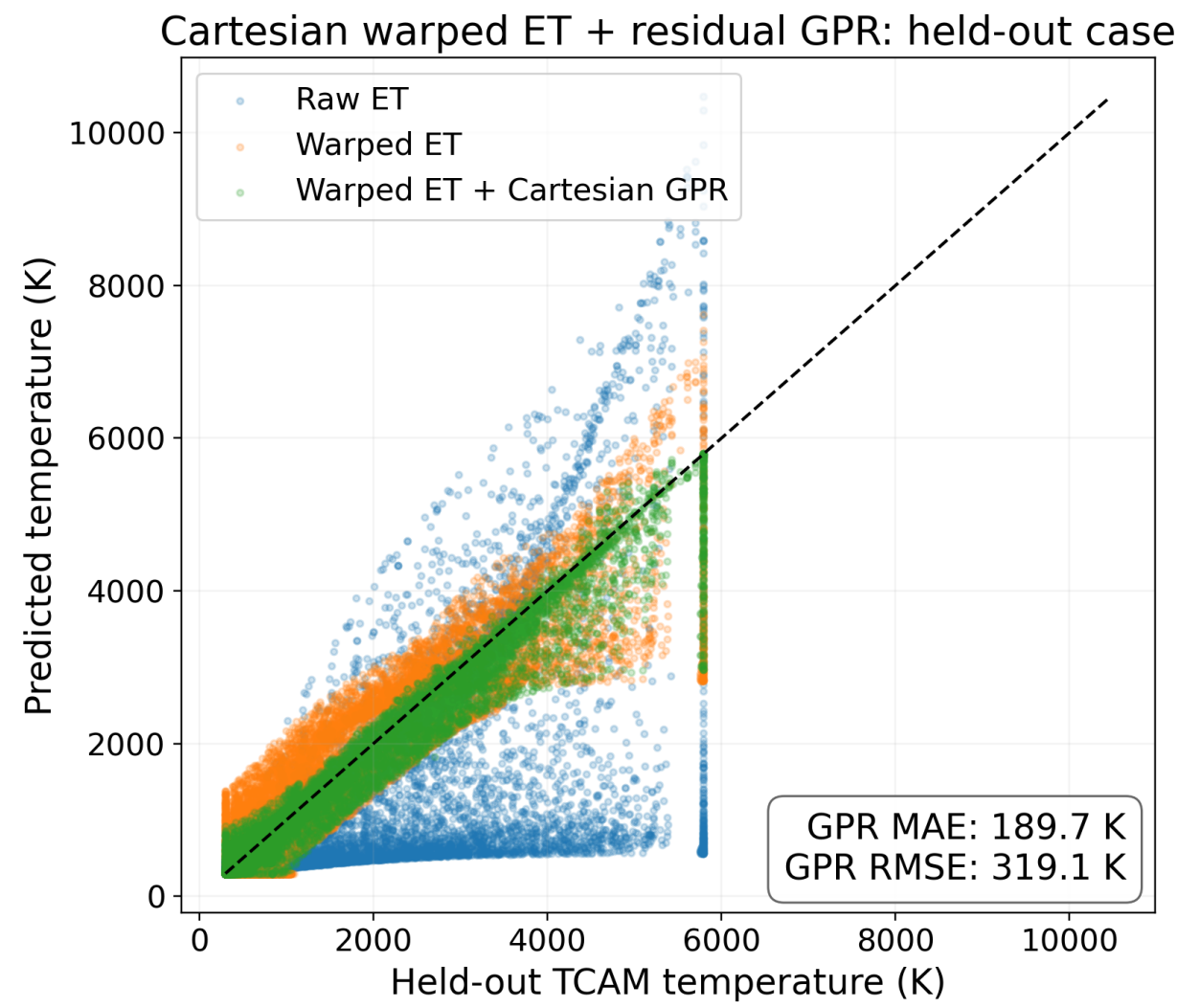
Original ET (left) and predicted warped ET (right), shown from the top and side.

Held-Out Melt Pool Prediction

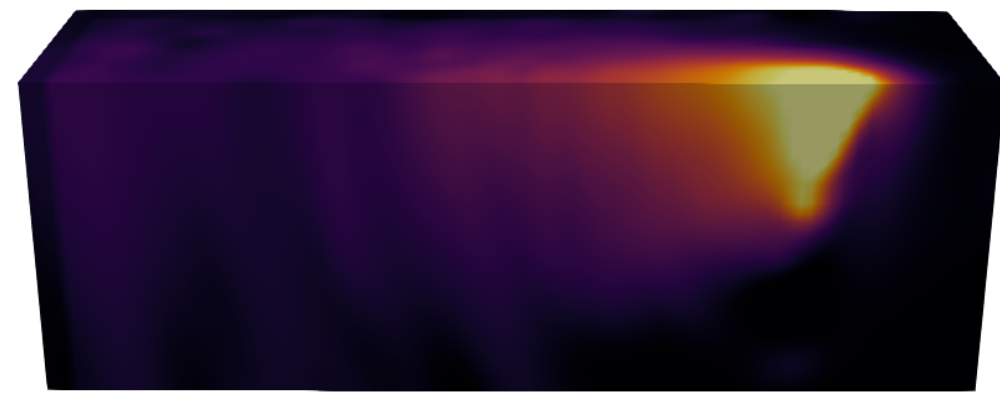
The 250 W, 0.5 m/s TCAM field is excluded from training and used only after prediction for evaluation.



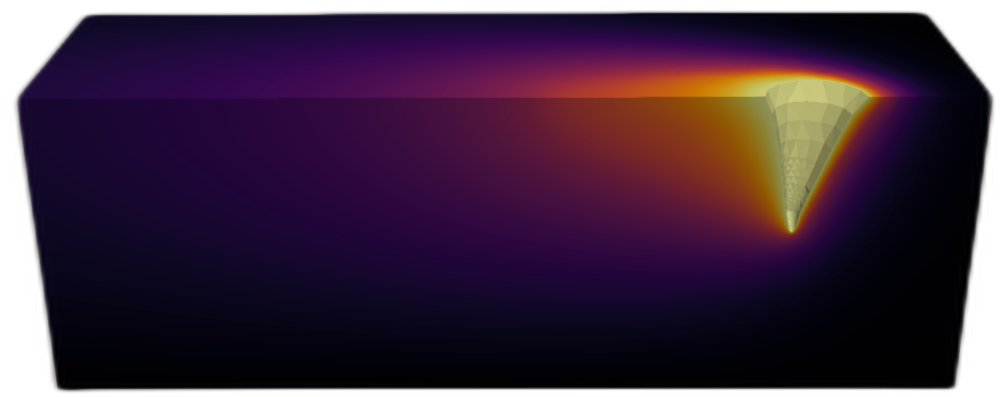
Blind Prediction vs. TCAM Ground Truth



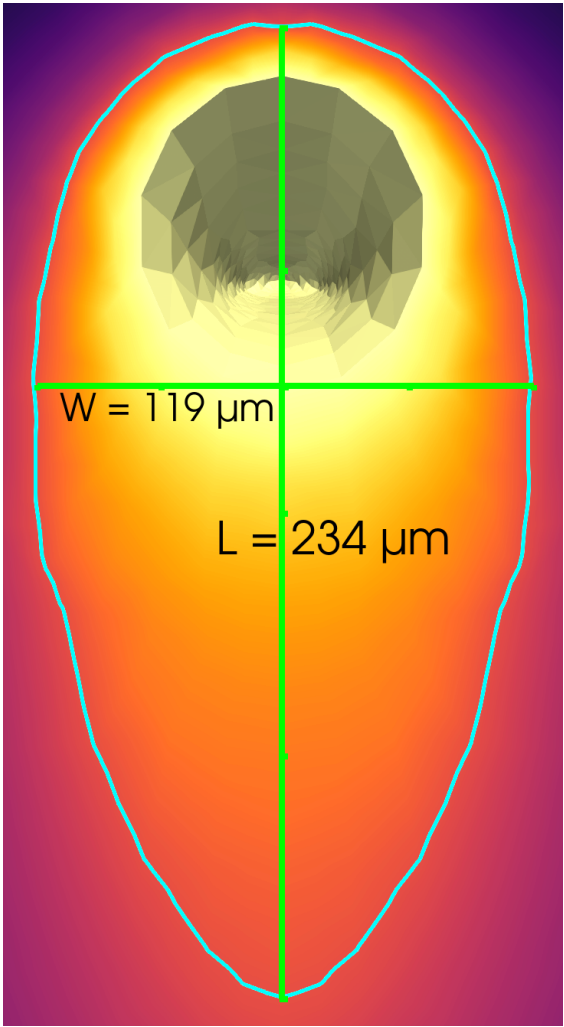
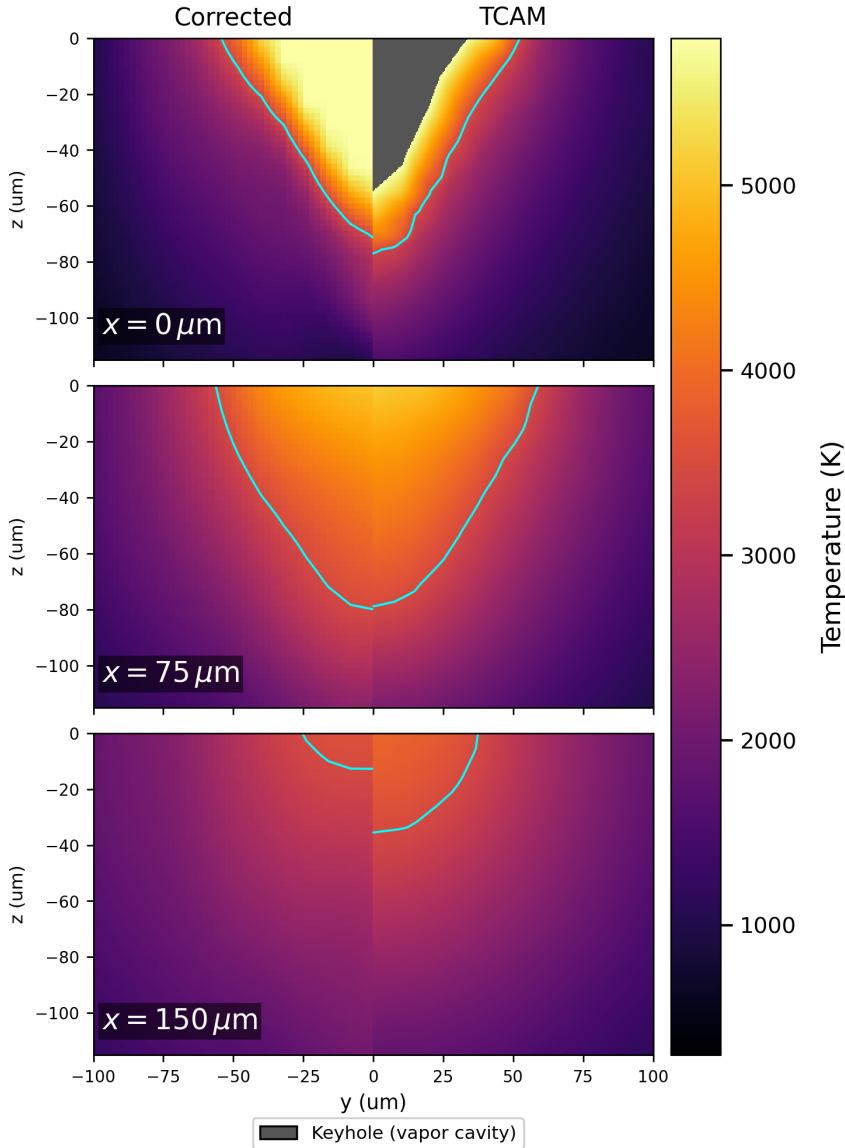
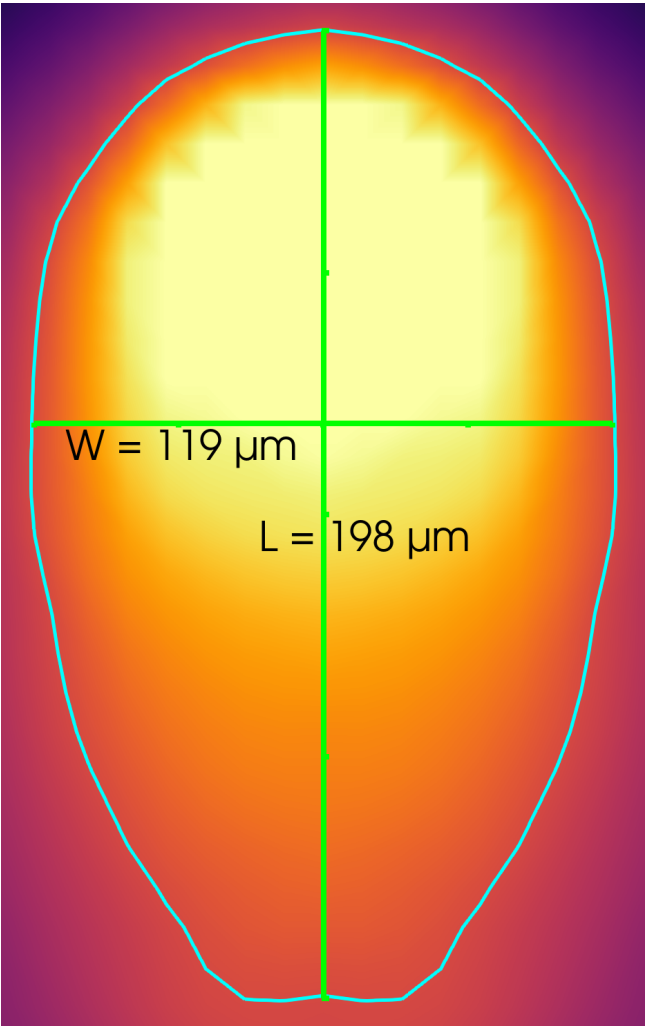
Blind Bayesian prediction



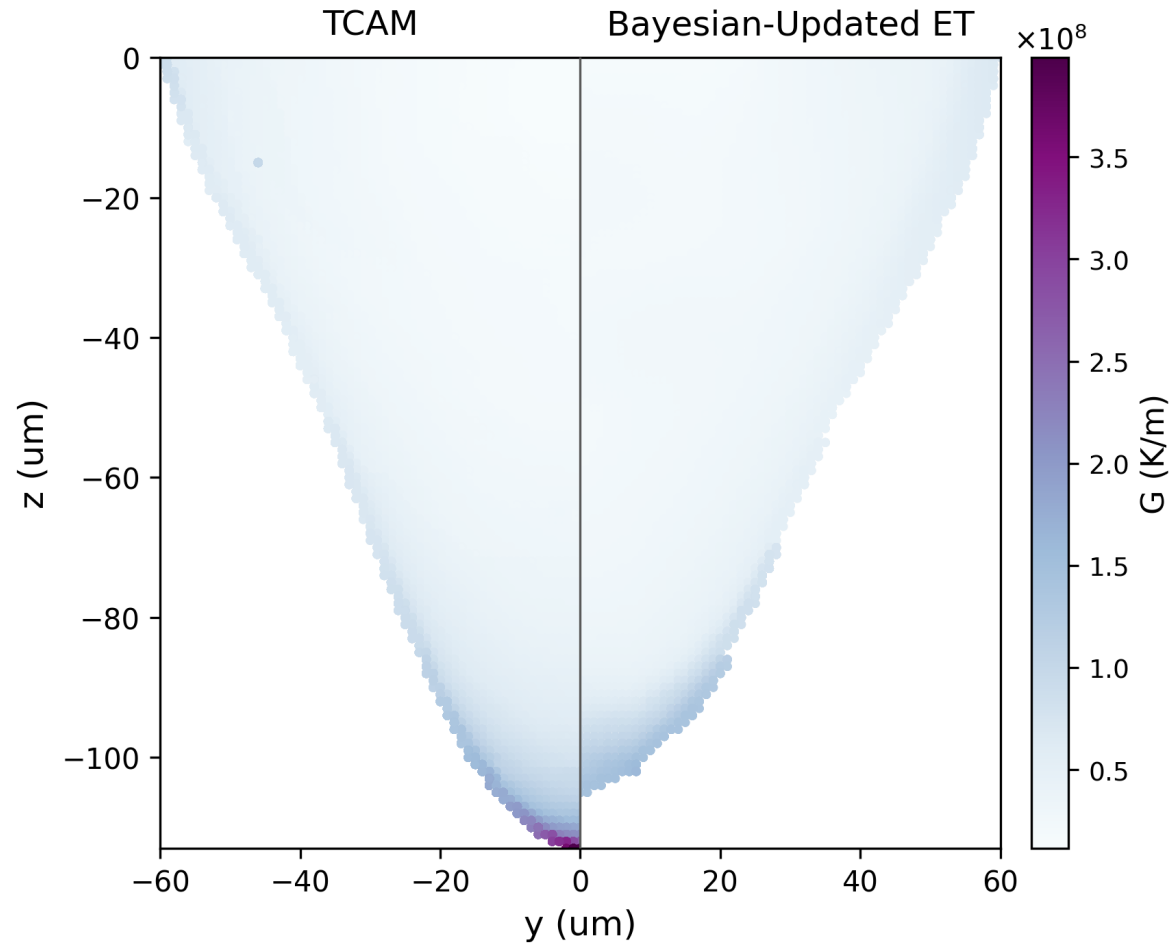
TCAM ground truth



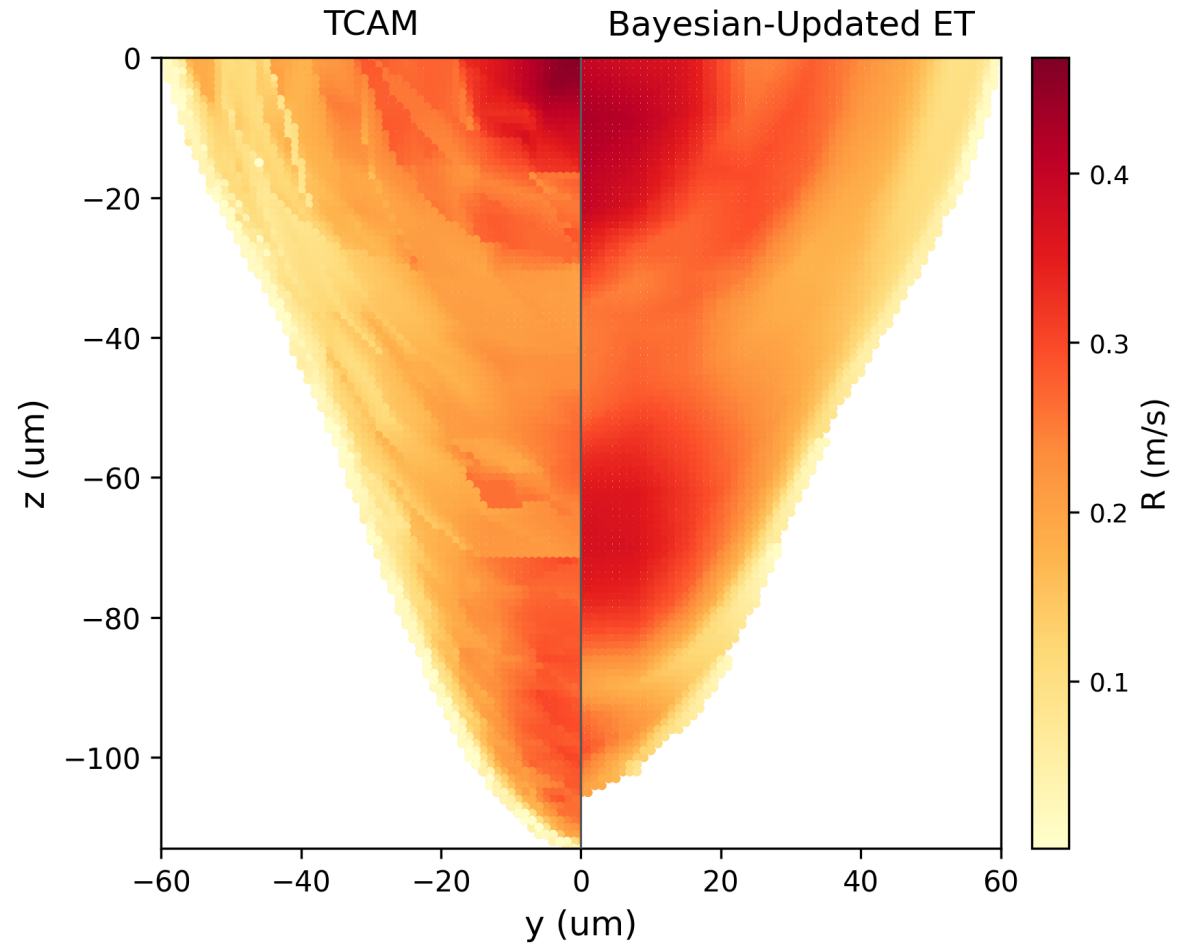
Held-Out Temperature Cross Sections



Held-Out Thermal Gradient and Solidification Rate

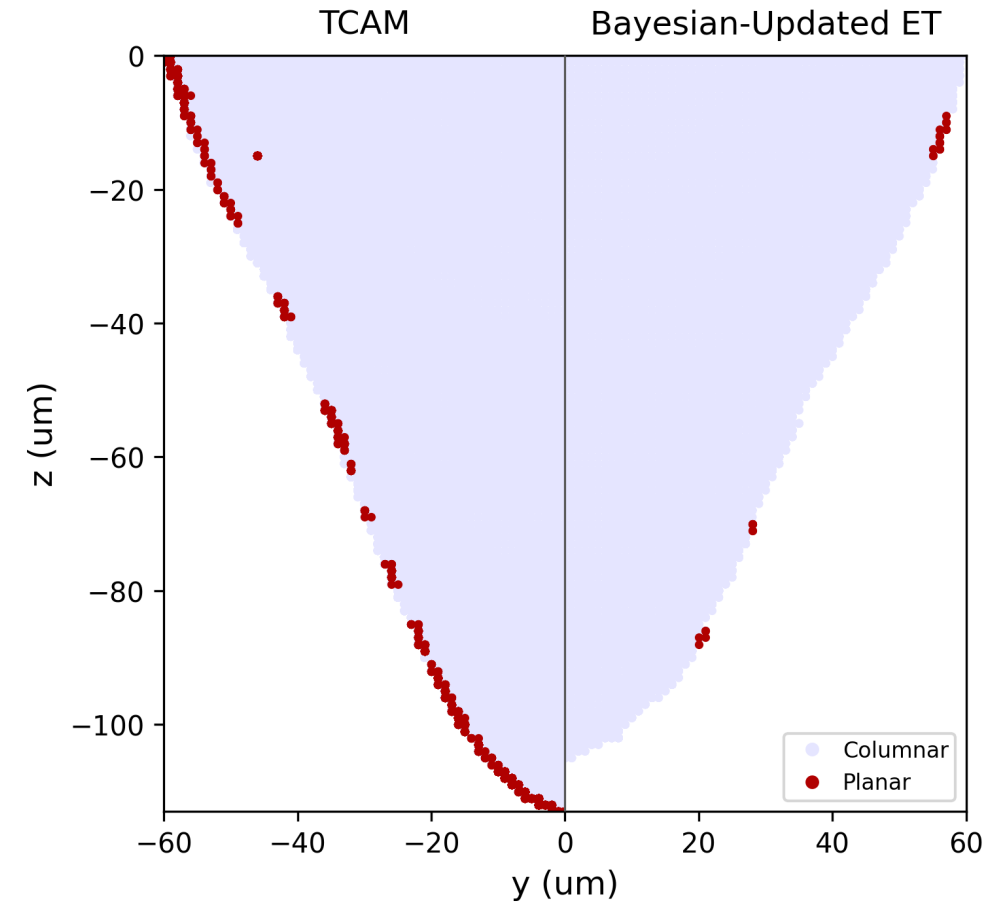
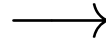
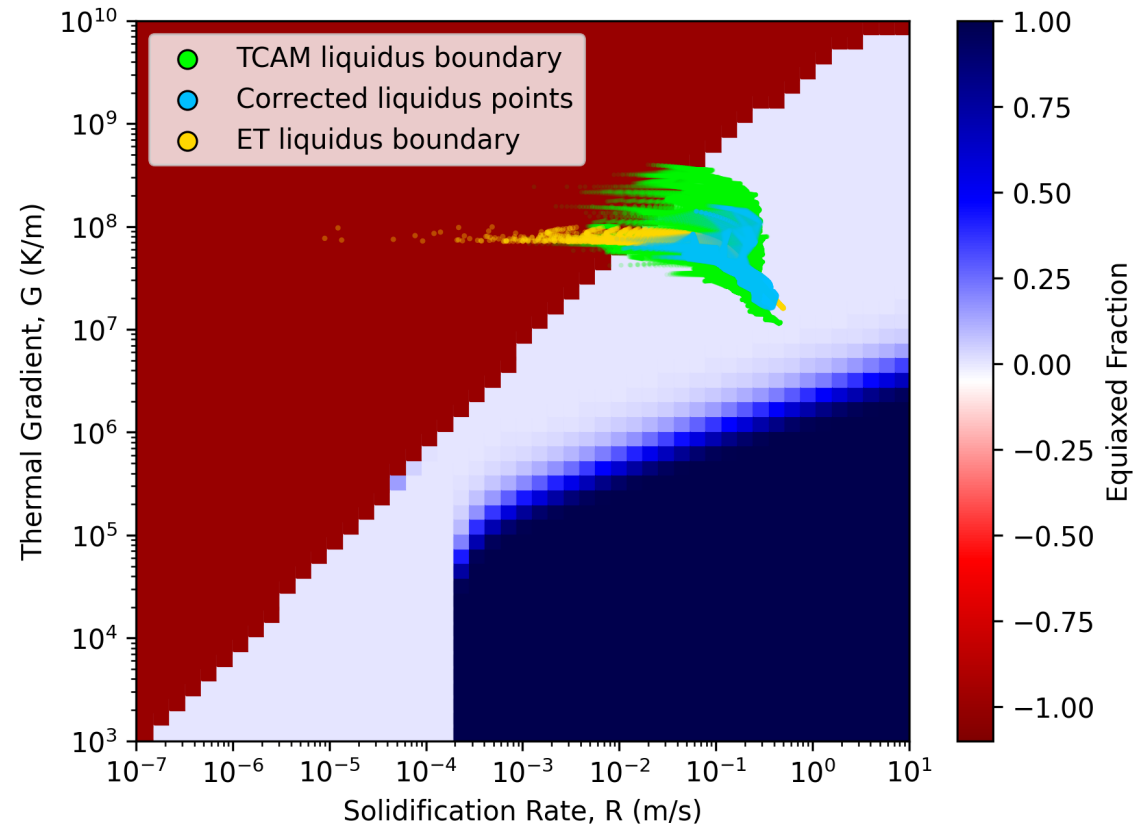


Thermal gradient G along the liquidus solidification front projected onto a yz slice: TCAM (left) and held-out Bayesian prediction (right).



Solidification rate R along the liquidus solidification front projected onto a yz slice: TCAM (left) and held-out Bayesian prediction (right).

Held-Out Microstructure Prediction



Microstructure classification along the liquidus solidification front projected onto a yz slice: TCAM (left) and held-out Bayesian prediction (right) for the same alloy and process condition.

Thermo-Calc CET G - R map with TCAM, held-out Bayesian prediction, and original ET liquidus-boundary conditions for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$, 250 W and 0.5 m/s.



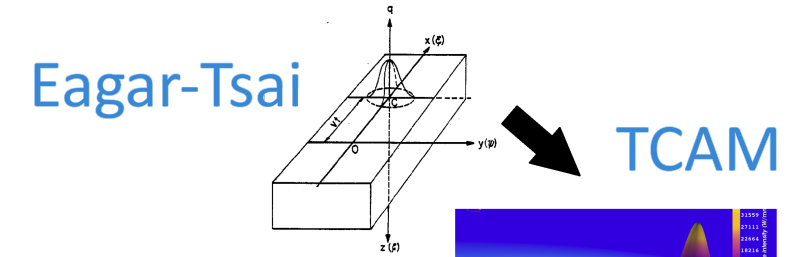
Closing

Accomplishments

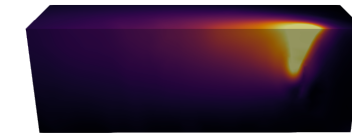
- Created an end-to-end **ET** → **KGT** workflow for rapid LPBF microstructure prediction.
- Used Bayesian updating to add **TCAM-informed fidelity** while retaining ET's low computational cost.
- Enabled **high-throughput process screening** from process parameters through thermal history to morphology.

Future Work

- Improve warped-GPR agreement on the G - R map.
- Test additional alloy systems.
- Predict the onset of keyholing.

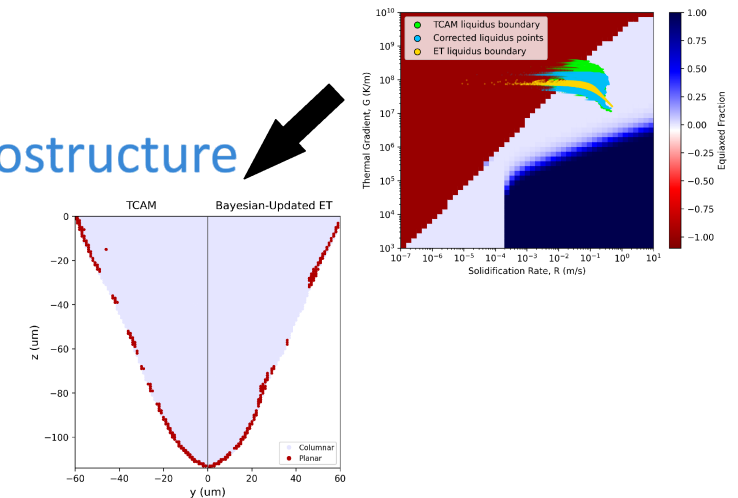


Bayesian-Updated
ET



G-R Map

Microstructure



Same morphology $\neq$ same underlying physics.

